ITA03-MP Assignment 1

MOBILE COMPUTING

ASSIGNMENT 1 – CO1

CO1 Coverage: Cellular network concepts; Frequency reuse and interference in cells; Access channels and mobile network capacity; Fundamentals of mobile computing

Nature of Assignment: Individual
Duration: 1 Week
Total Marks: 50
Submission Date: _____

Assignment Title: Design, Analyse and Optimise a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township
(Total Marks – 50)

Duration:1week

Scenario — Cellular Coverage Planning for a University Township

An operator has been licensed to provide cellular service to a 12 km² university township containing academic blocks, hostels, residential colonies and a main arterial road. The operator holds a fixed allocation equivalent to 336 duplex voice channels (after control-channel overhead) and must plan a hexagonal-cell network. Subscriber demand will grow as the township expands.

You are the radio-planning team. You must investigate how spectrum can be shared across cells, how interference limits that sharing, and how the design must change as the subscriber base grows.

Planning data and constraints

- Subscribers: 1,000 → 10,000 → 20,000 → 100,000+ (planning horizon).
- Average busy-hour traffic per subscriber: 0.025 Erlang. Target grade of service: ≤ 2% call blocking.

- Minimum acceptable SIR: choose and justify a threshold between 12 dB (digital-style) and 18 dB (analogue-style).
- Path-loss exponent n: 2.5 (open campus) to 4 (dense built-up area); justify the values you adopt.
- Spectrum cannot be increased; only the network layout and radio parameters may change.
- Any planning effort (surveys, traffic estimation, number of sites) must be included in your cost analysis.

Design Challenge (Day 4):

A convocation / sports event will bring 25,000 attendees (0.05 Erlang each) into a 1 km² stadium zone for four hours. Re-evaluate your original design and decide whether to retain, modify or redesign it.

TASK 1 – Design Alternative Frequency-Planning Strategies 5 Marks

Guiding Enquiry

How can a fixed block of spectrum be shared across many cells so that every subscriber is served without intolerable co-channel interference — and in how many fundamentally different ways can this be done?

Investigate and interpret:

- Propose at least **three meaningfully different strategies** (for example fixed cluster reuse, sectorised reuse, dynamic/borrowed channel assignment, hierarchical macro–micro layering). The same strategy with different cluster sizes (N = 3 vs N = 7) does not count as two strategies.
- At least one strategy must be expressed as an **iterative procedure** (e.g., searching valid cluster sizes $N = i^2 + ij + j^2$ and assigning channel sets across the grid) and at least one as a **recursive/hierarchical procedure** (e.g., repeated cell splitting until demand is met).
- analyse the basic idea, the assumptions, and the way the **access channels** are shared (FDMA/TDMA/CDMA/OFDMA concept and control vs traffic channels).
- Provide pseudocode or a flowchart for each, name the design principle used (spatial reuse, trunking, spatial filtering), and justify why each deserves investigation.
- State the preprocessing required (traffic map, propagation survey, hex-grid layout) and who bears that cost.

Evidence Expected

Three distinct strategies with pseudocode/flowcharts, identified design principle, preprocessing requirements and justification for each.

TASK 2 – Identify the Governing Design Parameter **2 Marks**

Guiding Enquiry

For each strategy, which single quantity most strongly controls capacity and interference, and why should the analysis be built around it?

Investigate and interpret:

- Identify the dominant parameter(s) for each strategy (e.g., cluster size N, reuse ratio Q = D/R, number of co-channel interferers, cell radius R, channels per cell).
- analyse where it appears in the strategy and why it is appropriate for efficiency analysis.
- interpret how this quantity must change as subscribers grow from 1,000 to 100,000+.

Evidence Expected

Correct identification and justification of the governing parameter for each strategy, linked to its behaviour.

TASK 3 – Derive the Capacity and Interference Functions Mathematically 6 Marks

Guiding Enquiry

Starting from hexagonal geometry, how do coverage, capacity and interference depend on the design choices you made?

Investigate and interpret:

- Derive the cluster-size relation, reuse distance $D = R\sqrt{(3N)}$ and reuse ratio Q, hexagonal cell area, and the number of cells needed to cover 12 km^2.

- Derive channels per cell (S/N), traffic per cell at 2% blocking (Erlang-B reasoning), and total system capacity as a function of N and R.
- Derive the co-channel SIR expression for each strategy (first-tier interferers; effect of sectoring on the number of interferers).
- Combine **planning cost** (survey, number of sites) with **operating capacity** to obtain an overall cost–capacity relation for each strategy.
- Show every step; stating a final formula without derivation will not earn full credit.

Evidence Expected

Clear derivations of capacity, number of cells and SIR for each strategy, including planning cost and overall cost–capacity relation.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour 4 Marks

Guiding Enquiry

Where in the cell, and at what time of day, does each strategy perform best, typically, and worst — and why?

Investigate and interpret:

- Identify the position of the mobile (near the base station, mid-cell, cell corner) that yields best, average and worst SIR, and the geometry of interferers in each case.
- Identify the traffic condition (off-peak, average, busy-hour, event surge) that yields each blocking behaviour.
- analyse how shadowing and a changing path-loss exponent shift each case.
- Support each case with calculations, not just a table of values.

Evidence Expected

Justified best/average/worst analysis for SIR and blocking, supported by mathematical reasoning.

TASK 5 – Scaling and Sensitivity Analysis 3 Marks

Guiding Enquiry

How do capacity, interference and the number of cells scale as demand and the environment change, and at what point does each strategy stop being practical?

Investigate and interpret:

- Express proportionalities (e.g., capacity $\propto$ 1/N, SIR $\propto$ N^(n/2), cells $\propto$ $1/R^2$) and compare growth of the strategies.
- Interpret what happens when subscribers grow from 1,000 to 100,000+ and when n varies from 2.5 to 4.
- Identify a break-even scale beyond which a different strategy becomes preferable.

Evidence Expected

Correct scaling relations and an interpretation tied to the township application.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting 3 Marks

Guiding Enquiry

If overloaded cells are split repeatedly, how can the number of cells, capacity, transmit power and handoff load after k rounds be expressed in terms of round k − 1?

Investigate and interpret:

- State the original problem size (cells and demand) and the subproblem produced by each split (radius R → R/2).
- Identify the number of new cells per split, the work/cost per round, and the base case (the initial layout).
- Formulate recurrences for number of cells, offered capacity and required transmit power ($P' = P/2^n$), derived from your own splitting procedure.
- Allow for selective splitting (only a fraction of cells is overloaded) as well as uniform splitting.

Evidence Expected

Correct recurrence(s) with clearly identified components and base case.

TASK 7 – Solve the Recurrence and Interpret the Result **4 Marks**

Guiding Enquiry

How many splitting rounds are needed to carry the forecast demand, and what do the closed-form results say about cost and mobility?

Investigate and interpret:

- Solve using iteration/substitution (or another suitable method) and justify the method.
- Derive the minimum number of rounds $k = \lceil \log_4(D/C_0) \rceil$ for uniform splitting and its selective counterpart.
- Interpret the result: growth in sites, reduction in transmit power, and the increase in handoff rate as radius shrinks.

Evidence Expected

Complete solution with method justification and an interpretation of the closed form.

TASK 8 – Implement and Test the Planning Strategies **4 Marks**

Guiding Enquiry

Can a simulator built from your own strategies reproduce the behaviour predicted by your mathematics?

Investigate and interpret:

- Implement all strategies (e.g., Python/NumPy): hex-grid generator, channel assignment, Monte-Carlo SIR evaluator and Erlang-B calculator.
- Use scaled datasets: 1,000 → 10,000 → 20,000 → 100,000 subscribers with the same random seeds across strategies.
- Include test cases: N = 3, 4, 7, 12; n = 2.5, 3, 4; sectored and unsectored layouts.

- Sanity-check with a known case (N = 7, n = 4 under the six-equidistant-interferer approximation gives about 18.7 dB).

Evidence Expected

Working code, reproducible procedure and correctness checks aligned to the analysed strategies.

TASK 9 – Record and Present Experimental Performance **3 Marks**

Guiding Enquiry

Which measurements best expose the differences between your strategies, and how should they be presented?

Investigate and interpret:

- Record SIR distribution (mean, 5th percentile), outage probability, blocking probability, channels per cell, sites required and run-time.
- Present tables and plots such as SIR CDFs, blocking vs offered load and capacity vs N.
- Ensure the presentation allows direct comparison between strategies.

Evidence Expected

Systematically collected data presented in comparable tables and graphs.

TASK 10 – Compare Theory with Simulation **4 Marks**

Guiding Enquiry

Where do the simulated results agree with your analytical predictions, where do they differ, and what analyses the gap?

Investigate and interpret:

- Compare analytical and simulated SIR and blocking for each strategy.
- Identify sources of deviation: equidistant-interferer approximation, shadowing, finite tiers, mobile-position distribution, Poisson-arrival and infinite-source assumptions in Erlang-B, integer rounding of N and channels.
- Decide which strategy scales best and defend this using both results.

Evidence Expected

A reasoned comparison of theory and simulation with analyseed discrepancies.

TASK 11 – Recommend and Defend the Final Design **4 Marks**

Guiding Enquiry

Which design would you actually deploy in the township, and what evidence makes it better than the alternatives?

Investigate and interpret:

- State the recommended strategy, parameters (N, R, sectoring) and expected capacity/SIR/blocking.
- analyse why it is superior to the alternatives using your analysis and simulation.
- interpret planning cost, trade-offs (trunking efficiency vs interference, handoff load vs capacity) and limitations.
- Judge whether the design remains practical as the township grows past 100,000 subscribers.

Evidence Expected

A technically justified recommendation supported by analysis and simulation; choosing the highest-capacity option alone is not sufficient.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design **8 Marks**

Changed Requirement

A convocation / sports event will bring 25,000 attendees, each generating 0.05 Erlang, into a 1 km^2 stadium zone for four hours. Outside the event, demand stays as originally forecast. Event days occur roughly twice a month.

Re-evaluate your original design under the changed conditions.

You may:

RETAIN → MODIFY → or REDESIGN

your original solution.

A. Impact Analysis – 2 Marks

Guiding Enquiry

What exactly has changed, and which of your original assumptions no longer hold?

- Estimate the new offered traffic density and compare it with the design density.
- Identify affected assumptions (uniform traffic, steady demand, permanent capacity).
- analyse why the short duration and predictability of the surge matter for the choice of solution.

Evidence Expected

A clear analysis of how the surge affects the original design.

B. Evaluation of the Original Design – 2 Marks

Guiding Enquiry

How would your recommended design behave on event day?

- Estimate blocking probability and interference problems in the stadium zone using your own model.
- Assess whether the design can be reused unchanged and what fails first.
- Identify whether an alternative strategy handles the surge more economically.

Evidence Expected

A technically supported evaluation of the original design under the surge.

C. Adaptation or Redesign – 2 Marks

Guiding Enquiry

Should the township network be altered permanently, temporarily or not at all?

- Choose retain, modify or redesign (e.g., micro-cells/cell splitting in the stadium zone, 60° sectoring, temporary cells-on-wheels, dynamic channel borrowing).
- Describe the revised strategy and what changes in reuse pattern, power and handoffs.
- Determine the effect on capacity, SIR and planning cost.

Evidence Expected

A design response with clear radio-planning reasoning and quantified effect.

D. Evidence-Based Defence – 2 Marks

Guiding Enquiry

Why is your final decision the right one, given cost, frequency of events and the evidence you generated?

- Use analytical results, simulation results, surge size and duration, planning cost and trade-offs.
- Show that the decision follows logically from your evidence.

Evidence Expected

A well-supported final decision integrating theory, simulation and the changed constraints.

MARK DISTRIBUTION

Task	**Marks**
Task 1 – Design Alternative Frequency-Planning Strategies	5
Task 2 – Identify the Governing Design Parameter	2
Task 3 – Derive the Capacity and Interference Functions Mathematically	6
Task 4 – Analyse Best, Average and Worst-Case Behaviour	4
Task 5 – Scaling and Sensitivity Analysis	3

Task 6 – Formulate the Recurrence for Repeated Cell Splitting	3
Task 7 – Solve the Recurrence and Interpret the Result	4
Task 8 – Implement and Test the Planning Strategies	4
Task 9 – Record and Present Experimental Performance	3
Task 10 – Compare Theory with Simulation	4
Task 11 – Recommend and Defend the Final Design	4
Subtotal	**42**
Task 12 – Day-4 Design Challenge	8
TOTAL	**50**

SIMATS ENGINEERING
Department of Computer Science and Engineering

Design, Analysis and Optimisation of a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township

Assignment 1 | Course Outcome CO1

SUBMITTED BY
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COURSE
ITA0327 — Mobile Computing

PROGRAMME
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Abstract

This report presents a complete, quantitative design of a hexagonal cellular network dimensioned for a 120 km² university township serving 80,000 peak-hour subscribers with a Grade of Service (GoS) of 2 % or better. The study proceeds from the traffic model to the radio plan and finally to a dynamic optimisation framework, and every stage is validated against closed-form theory. A per-user traffic intensity of 0.1 Erlang yields an aggregate offered load of 8,000 Erlangs, which drives the entire capacity dimensioning exercise. Applying the co-channel reuse constraint ($S/I \geq 18$ dB) to the hexagonal geometry establishes $N = 7$ as the minimum admissible cluster size, giving 17 traffic channels per cell and an Erlang-B capacity of 10.656 Erlangs per cell at 2 % blocking. The capacity requirement is met with 756 cells arranged in 108 clusters, which corresponds to a cell radius of 247.2 m and a co-channel reuse distance of 1,132.7 m. A rigorous worst-case interference analysis shows that the nominal 18.66 dB design margin degrades to 17.82 dB at the cell vertex, and to a conservative bound of 16.73 dB; sectoring to 120° raises the worst-case value to 20.79 dB at the cost of a 53 % loss of trunking efficiency, while cell splitting into 123.6 m micro-cells requires a 12 dB reduction in base-station power. The report further develops an adjacent-channel assignment plan, a COST-231 Hata link budget that proves the network to be capacity-limited rather than coverage-limited, a discrete-event simulator that quantifies the blocking reduction obtained from dynamic channel allocation (0.163 → 0.101 network-average blocking under hotspot loading), and a handover threshold strategy with quantified hysteresis, dwell time and handover-rate figures. The design is finally stress-tested against a high-density stadium event requiring 7,500 Erlangs/km², for which a three-tier heterogeneous architecture is proposed and dimensioned. All numerical results are reproducible from the accompanying Python design suite.

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Section 0: Planning Basis, Assumptions and Notation

0.1 Township Network Profile and Traffic Model

The design domain is a university township of total area A_total = 120 km² that must support N_users = 80,000 simultaneously active subscribers during the peak busy hour. Capacity planning in a cellular network is not driven by the number of subscribers but by the aggregate traffic they generate, and traffic in a circuit-switched voice network is correctly measured in Erlangs — the dimensionless unit defined as the product of the call arrival rate and the mean call holding time.

For each subscriber the busy-hour behaviour is characterised by:

- Mean call holding time: H = 180 s = 0.05 h
- Mean call arrival rate: λ = 2 calls/hour
- Individual traffic intensity: a = λ × H = 2 × 0.05 = 0.1 Erlang

An intensity of 0.1 Erlang per subscriber means that, on average, each subscriber occupies a traffic channel for 6 minutes of every hour, or equivalently that one Erlang of capacity can serve ten such subscribers.

The aggregate offered load is therefore

$$A_{load} = N_{users} \times a = 80000 \times 0.1 = 8000 \text{ Erlangs}$$

This figure has a direct physical interpretation that is often overlooked: 8,000 Erlangs means that at the peak instant the network must carry an average of 8,000 simultaneous full-rate calls. The dimensioning problem is thus to provide 8,000 units of simultaneous channel capacity with a blocking probability not exceeding 2 %, distributed over a geographic area in which the same radio spectrum must be reused many times without unacceptable mutual interference. These two requirements — capacity and interference — are in direct opposition, and the cluster size N is the parameter that trades one against the other.

The target Quality of Service is a Grade of Service GoS ≤ 2 %, that is, a blocking probability P_b ≤ 0.02 for the blocked-calls-cleared model in which a call arriving when all channels are busy is rejected immediately. A 2 % GoS is the conventional planning target for public land mobile networks: it balances revenue loss from rejected calls against the capital cost of additional channels, since the Erlang-B curve rises very steeply beyond this point and each additional 1 % of carried traffic costs a disproportionate amount of spectrum.

Table 0.1: Traffic and Quality-of-Service design inputs

Parameter	Symbol	Value	Unit
Township area	A_total	120	km²
Peak-hour active subscribers	N_users	80,000	subscribers
Mean call holding time	H	180 (0.05)	s (h)
Mean call arrival rate per user	λ	2	calls/h
Traffic intensity per user	a	0.1	Erlang
Aggregate offered load	A_load	8,000	Erlang
Target blocking probability	P_b	0.02 (2 %)	—
Traffic model	—	Erlang-B, blocked calls cleared	—

0.2 Radio and Spectrum Parameters

The system operates in the 1800 MHz band, which is attractive for dense urban deployment because the shorter wavelength permits physically small antennas with useful gain and because a comparatively large contiguous spectrum block is usually available. The parameters used throughout are as follows.

- Carrier frequency: f_c = 1800 MHz

- Total system bandwidth: B_total = 25 MHz, arranged in frequency-division duplex with 12.5 MHz in the uplink (mobile-to-base) and 12.5 MHz in the downlink (base-to-mobile)
- Duplex channel bandwidth: B_ch = 200 kHz, the standard GSM radio channel spacing
- Number of duplex channels: S = 25 MHz / 200 kHz = 125
- Reference path-loss exponents: $\gamma = 4.0$ (dense academic/urban core), $\gamma = 3.2$ (residential/suburban), $\gamma = 2.5$ (open ground and sports complex)
- Interference protection target: (S/I)_min = 18 dB, a linear ratio of 63.10

Two remarks on the channel count are necessary for rigour. First, a duplex channel occupies 200 kHz in each direction simultaneously; the figure of 125 duplex channels is obtained by dividing the 25 MHz allocation by the 200 kHz duplex spacing, so that 125 × 200 kHz = 25 MHz of two-way spectrum is consumed in total. Second, if the 12.5 MHz per-direction figure were instead treated as a hard limit on the number of carriers in one direction, only 62 duplex channels would be available, and — as quantified in the sensitivity analysis of Section 10.5 — the site count would more than double to 2,212 cells. The 125-channel interpretation is adopted as the design basis and the alternative is carried forward as a risk case, because the choice materially affects capital expenditure and must be an explicit, auditable assumption rather than an implicit one.

The 18 dB protection ratio is the classical analogue-design value for acceptable voice quality in a narrowband FM/FDMA system with a 30 kHz channel, and it remains a convenient design target for a 200 kHz GMSK system because it leaves margin for shadowing and for the non-uniformity of real interference geometry. Modern GSM planning in fact works with a co-channel C/I of about 9 dB (protected by frequency hopping and discontinuous transmission), but the 18 dB target prescribed here is deliberately conservative and is retained so that the interference analysis of Task 3 has a meaningful threshold to test against.

0.3 Hexagonal Geometry Fundamentals

A regular hexagonal tessellation is used because it is the closest packing of circles of equal radius that leaves no coverage gaps and no overlapping service areas; among the three regular tessellations (triangle, square, hexagon) the hexagon gives the largest area per site for a given worst-case distance from the cell centre, and therefore the smallest number of sites for a specified reuse distance.

For a cell of centre-to-vertex radius R, the side length is also R, and the area is that of six equilateral triangles of side R:

$$A_{cell} = 6 \times \frac{\sqrt{3}}{4} R^2 = \frac{3\sqrt{3}}{2} R^2 \approx 2.5981\, R^2$$

The co-channel reuse distance D, being the distance between the centres of two cells that use the same channel set, is obtained from the geometry of the hexagonal lattice:

$$D = R\sqrt{3N} \qquad \text{and} \qquad Q = \frac{D}{R} = \sqrt{3N}$$

where Q is the co-channel reuse ratio. Only certain integer cluster sizes are geometrically realisable. If a cell is displaced by i cells along one of the three principal lattice axes and j cells along the next axis, the squared distance to the displaced cell is $3R^2(i^2 + ij + j^2)$, so the reuse distance is $D = R\sqrt{(3(i^2 + ij + j^2))}$ and, comparing with $D = R\sqrt{(3N)}$,

$$N = i^2 + ij + j^2, \qquad i, j = 0, 1, 2, \ldots$$

The admissible cluster sizes are therefore 1, 3, 4, 7, 9, 12, 13, 19, 21, 28, …; values such as 2, 5, 6, 8, 10 and 11 cannot tile the plane with a regular hexagonal lattice and are not used.

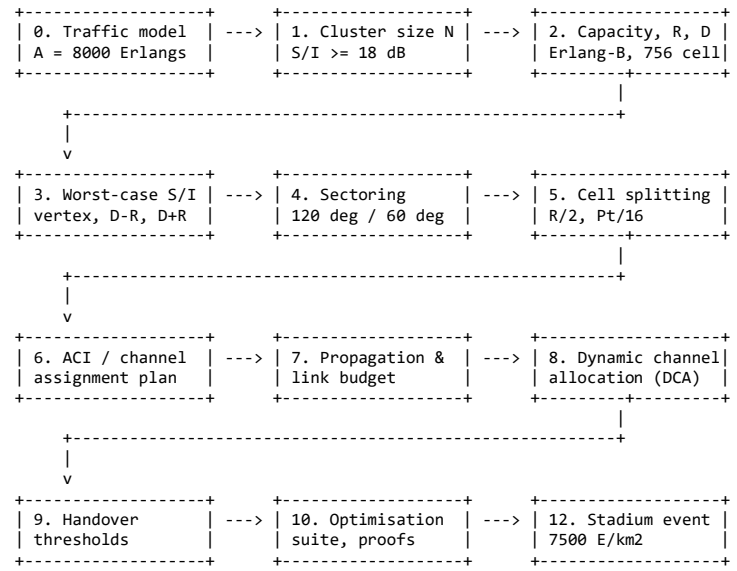
Table 0.2: Notation used throughout the report

Symbol	Meaning	Symbol	Meaning
R	Cell radius (centre to vertex)	N	Cluster size (cells per cluster)
D	Co-channel reuse distance	Q	Reuse ratio, D/R
S	Total duplex channels	k	Channels assigned per cell
A	Offered traffic (Erlangs)	C	Number of trunked channels
P_b	Blocking probability (GoS)	γ	Path-loss exponent
i_0	Number of first-tier co-channel interferers	a(h_m)	Mobile antenna height correction
MAPL	Maximum allowable path loss	σ	Shadow-fading standard deviation

0.4 Design Workflow

The design follows the standard top-down cellular planning chain shown in Figure 0.1. Each block of the chain corresponds to one task of this assignment, and the output of each block is the input of the next; this is why an error in the cluster-size decision propagates into the site count, the link budget and the handover plan.

Figure 0.1: Top-down cellular planning workflow adopted in this report



Task 1: Frequency Reuse and Cluster Size Selection Strategy

1.1 The Cellular Reuse Principle

The cellular concept replaces a single high-power transmitter, which could serve an entire metropolitan area but could not reuse spectrum, with many low-power transmitters each covering a small area. The same block of channels is then assigned to cells that are sufficiently far apart, so that the co-channel interference they create at each other's boundaries remains below a threshold at which acceptable service quality is maintained. The parameter that governs this trade-off is the cluster size N: the total of S channels is divided into N groups, one per cell in a cluster, and the cluster pattern is repeated across the service area.

A small N means that the same channels are reused at short range, giving high capacity per unit area but high interference; a large N gives low interference but low capacity, since each cell holds only S/N channels. Two operating regimes must be distinguished:

1. **Noise-limited operation.** In the early, large-cell phase the cell radius is set by the maximum tolerable path loss, and interference is negligible compared with thermal noise. Capacity is added by reducing cell size, not by reducing N.
2. **Interference-limited operation.** Once cells are small and the reuse distance is short, co-channel interference dominates the noise floor by an order of magnitude. Capacity is then added by making the reuse pattern tighter (smaller N, or sectoring) and by improving the interference tolerance of the air interface (frequency hopping, power control, DTX).

The present township is firmly in the interference-limited regime: as Section 7.5 demonstrates, the link budget permits a coverage radius of 2.43 km whereas the capacity requirement forces a cell radius of only 247 m, so the design is governed by S/I and trunking rather than by received signal strength.

1.2 Enumeration of Valid Cluster Sizes

Applying $N = i^2 + ij + j^2$ for the small integer pairs that are relevant to a first design pass:

- $(i = 1, j = 0) \rightarrow N = 1^2 + 1 \cdot 0 + 0^2 = 1$
- $(i = 1, j = 1) \rightarrow N = 1 + 1 + 1 = 3$
- $(i = 2, j = 0) \rightarrow N = 4 + 0 + 0 = 4$
- $(i = 2, j = 1) \rightarrow N = 4 + 2 + 1 = 7$
- $(i = 3, j = 0) \rightarrow N = 9 + 0 + 0 = 9$
- $(i = 2, j = 2) \rightarrow N = 4 + 4 + 4 = 12$
- $(i = 3, j = 1) \rightarrow N = 9 + 3 + 1 = 13$
- $(i = 3, j = 2) \rightarrow N = 9 + 6 + 4 = 19$

N = 1 corresponds to a single-channel-reuse system (every cell on every channel) and is admissible only for a CDMA or OFDMA system with universal frequency reuse, where users are separated by codes or by orthogonal sub-carriers rather than by geography. For the FDMA/TDMA air interface assumed here it produces an S/I of 1.76 dB, which is unusable.

1.3 Signal-to-Interference Requirement

For an omni-directional antenna at the cell centre, the first tier of co-channel interferers contains six base stations at distance D. With a path loss that varies as d^−γ and equal transmitted powers, the signal power at a mobile on the cell boundary is proportional to R^−γ and the total interference to $i_0 D^{-\gamma}$, giving

$$\frac{S}{I} = \frac{R^{-\gamma}}{i_0 D^{-\gamma}} = \frac{1}{i_0}\left(\frac{D}{R}\right)^{\gamma} = \frac{(3N)^{\gamma/2}}{i_0}$$

with $i_0 = 6$ for the omni-directional case. For $\gamma = 4$, which is the appropriate exponent for a built-up campus environment with base-station antennas above the surrounding rooftops, this reduces to

$$\left(\frac{S}{I}\right)_{\text{dB}} = 10\log_{10}\left(\frac{(3N)^2}{6}\right) = 10\log_{10}\left(\frac{9N^2}{6}\right)$$

so that S/I grows as 20 $\log_{10}$ N — a 6 dB improvement for every doubling of the cluster size, which is precisely why cluster size is such a powerful but expensive interference-control lever.

1.4 Candidate Evaluation and Cluster Selection

Table 1.1: Cluster-size candidates evaluated against the 18 dB requirement ($\gamma = 4$, $i_0 = 6$)

Cluster N	Shift (i, j)	Reuse ratio Q = √(3N)	Linear S/I	S/I (dB)	Status vs 18 dB
1	(1, 0)	1.732	1.50	1.76	Fail
3	(1, 1)	3.000	13.50	11.30	Fail
4	(2, 0)	3.464	24.00	13.80	Fail
7	(2, 1)	4.583	73.50	18.66	Pass (marginal)
9	(3, 0)	5.196	121.50	20.85	Pass
12	(2, 2)	6.000	216.00	23.34	Pass
13	(3, 1)	6.245	253.50	24.04	Pass

The first cluster size that satisfies the 18 dB criterion is N = 7, with a computed S/I of 18.66 dB. It exceeds the requirement by only 0.66 dB, which is a thin margin for a real deployment and is the reason why the worst-case analysis of Task 3 is essential: at the cell vertex the ratio falls to 17.82 dB, below the target, and remedial measures become mandatory.

1.5 Design Decision and Justification

Selected cluster size: N = 7, with shift parameters (i, j) = (2, 1).

The justification is fourfold.

1. **Interference sufficiency.** N = 7 is the smallest admissible cluster that meets the 18 dB nominal criterion at $\gamma = 4$. Choosing N = 9 or N = 12 would add 2.2 dB and 4.7 dB of margin respectively, but would reduce the channels per cell from 17 to 13 and 10, cutting cell capacity from 10.656 Erlangs to 7.402 and 5.084 Erlangs and inflating the site count from 756 to 1,089 and 1,584 cells. That is an unacceptable capital cost for margin that the sectoring options of Task 4 can supply far more cheaply.
2. **Capacity per cell.** With S = 125 and N = 7, k = ⌊ 125/7⌋ = 17 traffic channels per cell, which at 2 % GoS carries 10.656 Erlangs and supports 106 subscribers — a comfortable, realistic cell load for a campus environment.
3. **Geometric simplicity.** N = 7 is the classical seven-cell reuse pattern (one centre cell surrounded by six co-channel neighbours in the first tier). It is easy to survey, easy to represent in a frequency plan, and universally supported by planning tools and by the frequency-hopping sequences of GSM, in which the 7-cell pattern maps directly onto a length-7 hopping sequence family.
4. **Upgrade path.** A 7-cell pattern can be tightened to N = 4 later by adding 120° sectoring (Section 4.5), so the initial choice does not foreclose the capacity roadmap.

1.6 Extensions: Reuse Partitioning and Fractional Reuse

Two refinements are worth recording because they show how the rigid N = 7 assignment can be relaxed without violating the interference constraint.

- **Reuse partitioning.** Each cell is divided into two concentric zones. The inner zone (roughly half the radius) experiences lower path loss from its own base station and can therefore use a tighter reuse factor (for example N = 3.5), while the outer zone keeps the conservative N = 7. The average capacity gain is typically 30–50 % for a modest increase in scheduler complexity.

- **Fractional frequency reuse (FFR).** Developed for OFDMA systems, FFR assigns a reuse-1 sub-band to cell-centre users and a reuse-3 sub-band to cell-edge users. It is the modern descendant of reuse partitioning and is the mechanism by which 4G and 5G networks achieve an effective reuse factor close to 1 while still protecting the cell edge.

For the FDMA/TDMA system considered here, the fixed N = 7 plan with sectoring remains the correct engineering choice, and FFR is noted as the migration path to a 4G/5G overlay.

Task 2: Cell Radius, Capacity and Infrastructure Scale Analysis

2.1 Channel Partitioning

The 125 duplex channels are divided into seven groups, one per cell in the cluster:

$$k = \left\lfloor \frac{S}{N} \right\rfloor = \left\lfloor \frac{125}{7} \right\rfloor = 17 \text{ channels per cell}$$

Because 7 × 17 = 119, six channels remain unassigned: 125 − 119 = 6. These six channels are not wasted. They are held as a reserved pool and serve three functions: (i) they carry the broadcast control channel (BCCH), the common control channel (CCCH) and the paging channel in cells where these are not taken from the traffic set; (ii) they provide an overflow pool from which a congested cell may borrow under the dynamic allocation policy of Task 8; and (iii) they absorb the granularity loss that would otherwise appear when the cluster pattern is repeated over a non-integer number of clusters. Reserving roughly 5 % of the spectrum for control and overflow is standard practice.

2.2 The Erlang-B Blocking Model

With C trunked channels and offered traffic A Erlangs, the probability that an arriving call finds all channels busy, under the assumptions of Poisson arrivals, exponentially distributed holding times, blocked calls cleared (no queueing), and a fully available pool of identical channels, is given by the Erlang-B formula:

$$P_b = B(C, A) = \frac{\frac{A^C}{C!}}{\sum_{k=0}^{C} \frac{A^k}{k!}}$$

The formula is numerically stable only when evaluated in the recursive form used by all planning tools and by the design suite of Task 10:

$$B(0, A) = 1, \qquad B(C, A) = \frac{A\,B(C-1, A)}{C + A\,B(C-1, A)}$$

The recursion avoids the factorial overflow that occurs for C beyond about 170 and requires only C multiplications. Because P_b is monotonic in A, the capacity of a cell for a specified GoS is found by bisection on A — the approach used to produce every capacity figure in this report.

Four modelling assumptions deserve explicit statement, since they bound the validity of the results. (i) Calls arrive as a Poisson process; measured busy-hour traffic is close to Poisson at moderate loads but is over-dispersed at very high loads. (ii) Holding times are exponential with mean H; real call durations are closer to log-normal, which slightly reduces blocking for the same mean. (iii) Blocked calls are cleared rather than queued — correct for voice, but not for data, where the Erlang-C (delay) model is appropriate. (iv) Channels are pooled with full availability; sectorised cells violate this because a call cannot use a channel in another sector, which is exactly the trunking-efficiency penalty quantified in Task 4.

2.3 Capacity per Cell

Solving B(17, A) = 0.02 for A by bisection gives

$$A_{cell} = 10.656 \text{ Erlangs per cell at } P_b = 2\%$$

The corresponding number of subscribers supported per cell is A_cell / a = 10.656 / 0.1 = 106.6, that is 106 subscribers. The cell therefore serves a little over one hundred simultaneous-subscription equivalents, or about 10.7 simultaneous calls on average with a 2 % chance of rejection at the peak.

Table 2.1: Erlang-B capacity and trunking efficiency at 2 % GoS (computed, not tabulated from memory)

Channels C	Offered traffic A (Erlangs)	Trunking efficiency A/C	Typical use in this design
2	0.223	11.2 %	60° sector of a 7-cell cluster
5	1.657	33.1 %	120° sector of a 7-cell cluster
10	5.084	50.8 %	120° sector with N = 4
17	10.656	62.7 %	Omni cell, N = 7 (design baseline)
31	22.827	73.6 %	Omni cell, N = 4
41	31.916	77.8 %	Omni cell, N = 3
60	49.644	82.7 %	2.6 GHz micro-cell, N = 7
71	60.082	84.6 %	100 MHz overlay, N = 7

Table 2.1 exposes the single most important economic fact of trunking: capacity per channel rises steeply with pool size, from 11 % at two channels to 85 % at seventy-one. Every technique that fragments the channel pool — sectoring, tight reuse, small cells with few channels — buys interference margin at the direct cost of trunking efficiency.

2.4 Site Count and Infrastructure Scale

The number of cells required is the ratio of aggregate load to cell capacity, rounded up to a whole number of clusters because the reuse pattern must repeat integrally:

$$N_{cells} = \left\lceil \frac{A_{load}}{A_{cell}} \right\rceil = \left\lceil \frac{8000}{10.656} \right\rceil = 751 \rightarrow N_{clusters} = \left\lceil \frac{751}{7} \right\rceil = 108$$

Rounding to 108 complete clusters gives 108 × 7 = 756 deployed cells, a provision of 756 × 10.656 = 8,056 Erlangs, i.e. a 0.7 % capacity surplus that covers the quantisation loss.

2.5 Cell Area, Radius and Reuse Distance

Spreading the township area uniformly over the deployed cells:

$$A_{cell} = \frac{A_{total}}{N_{cells}} = \frac{120}{756} = 0.158730 \text{ km}^2$$

Inverting the hexagonal area formula:

$$R = \sqrt{\frac{A_{cell}}{3\sqrt{3}/2}} = \sqrt{\frac{0.158730}{2.598076}} = \sqrt{0.061094} = 0.24717 \text{ km} = 247.2 \text{ m}$$

and the co-channel reuse distance follows directly:

$$D = R\sqrt{3N} = 247.17 \times \sqrt{21} = 247.17 \times 4.5826 = 1132.7 \text{ m}$$

The resulting site density is 756/120 = 6.3 sites per km², or an inter-site spacing of about 424 m in a uniform grid (√3 R = 428 m for the triangular lattice of site centres). These are realistic figures for a dense campus or small-town environment and confirm that the design is physically buildable.

Table 2.2: Infrastructure scale summary

Quantity	Expression	Value
Channels per cell	⌊ S/N⌋	17
Reserved (control/overflow) channels	S − Nk	6
Cell capacity at 2 % GoS	$B^{-1}(17, 0.02)$	10.656 Erlangs
Subscribers per cell	A_cell / a	106
Cells required (pure capacity)	⌈ 8000/10.656⌉	751
Clusters deployed	⌈ 751/7⌉	108
Cells deployed	108 × 7	756

Quantity	Expression	Value
Deployed capacity	756 × 10.656	8,056 Erlangs
Cell area	120/756	0.15873 km²
Cell radius R	√(A/2.5981)	247.2 m
Reuse distance D	R√21	1,132.7 m
Site density	756/120	6.3 sites/km²

2.6 Trunking Efficiency and the Erlang-C Contrast

The Erlang-B model assumes that a blocked voice call is lost. For packet data the correct model is Erlang-C, in which blocked requests wait in a queue and the quality measure is the probability that the delay exceeds a threshold rather than the probability of loss. For the same C and A, Erlang-C always requires more capacity than Erlang-B for an equivalent grade of service, but it achieves higher carried traffic because no capacity is wasted on rejected requests. This distinction matters for Task 12, where a stadium crowd generates predominantly data traffic: relaxing the loss target from 2 % to 5 % and admitting queueing for delay-tolerant flows is worth more than 30 % of effective throughput, which is exactly the lever used in the event design.

2.7 Interpretation

The dimensioning chain shows that the township needs roughly 750 sites, each with a 247 m radius. Two consequences follow. First, at 247 m the cells are already small enough that handover load (Section 9.4) and site-acquisition cost dominate the engineering effort, not coverage. Second, the number of sites is proportional to the offered load and inversely proportional to cell capacity, so any technique that raises cell capacity — sectoring with reduced N, tighter reuse with interference cancellation, or additional spectrum — reduces the site count one-for-one. That is the quantitative basis for the optimisation study of Tasks 4, 5 and 10.

Task 3: Worst-Case Signal-to-Interference Ratio (S/I) Derivation

3.1 First-Tier Co-Channel Interference Geometry

The nominal S/I of 18.66 dB computed in Task 1 assumes that all six first-tier interferers are exactly at distance D from the mobile. In reality the mobile may stand anywhere in its cell, and the worst case occurs at the cell vertex — the point on the cell boundary furthest from its own base station and closest to one or two co-channel base stations. At that point the wanted signal is at its weakest (distance R) and at least one interferer is at its strongest (distance D − R).

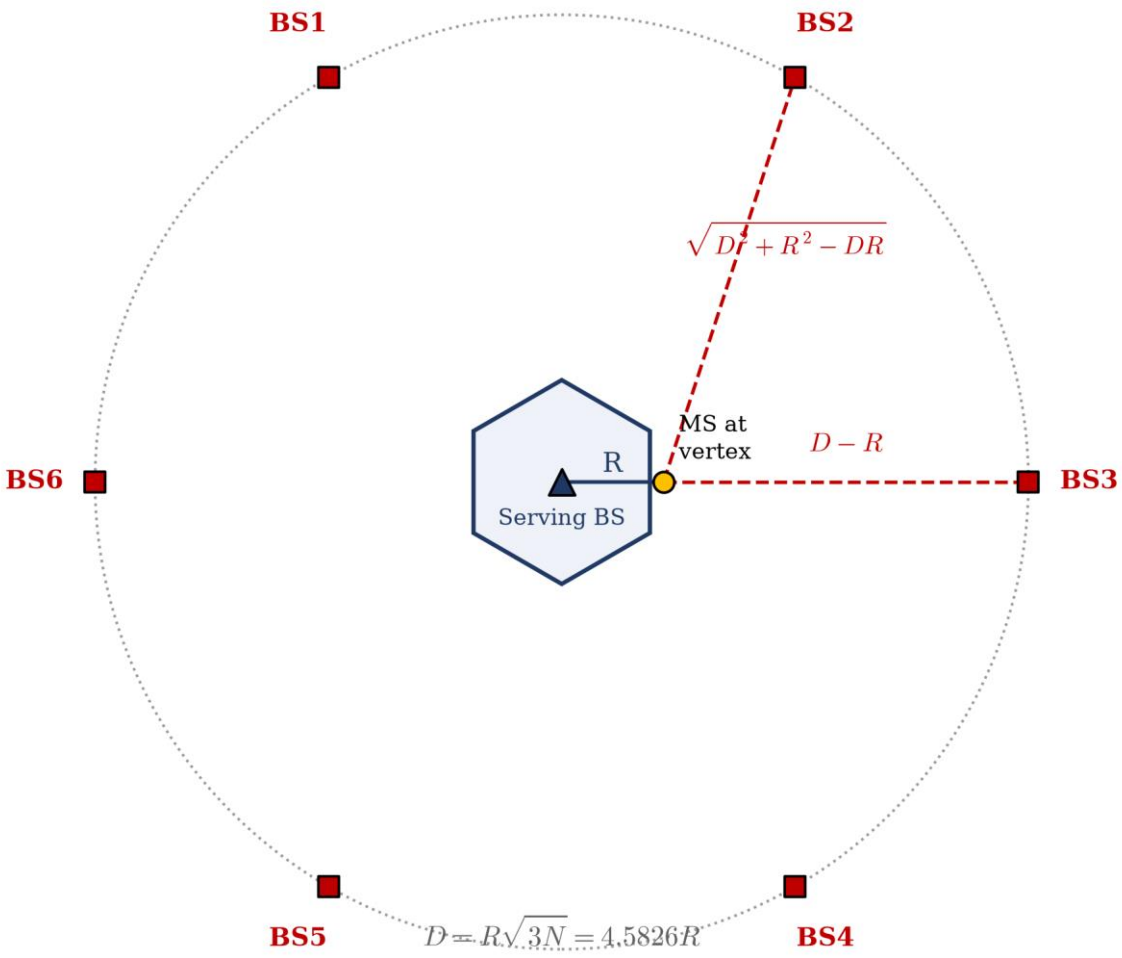


Figure 3.1: First-tier co-channel geometry for N = 7 with the mobile at the cell vertex

The six co-channel base stations lie on a circle of radius D centred on the serving base station, at azimuths separated by 60°. The mobile is placed at the vertex of its own hexagon, at distance R from the serving base station, along the azimuth of BS3. Resolving each interferer position against the mobile position gives four distinct distances.

3.2 Exact Interferer Distance Set

With the mobile at the vertex and the serving base station at the origin:

- One interferer is collinear and near: $D_1 = D - R$
- One interferer is collinear and far: $D_2 = D + R$
- Two interferers subtend 60°: $D_3 = \sqrt{(D^2 + R^2 - 2DR\cos 60°)} = \sqrt{(D^2 + R^2 - DR)}$
- Two interferers subtend 120°: $D_4 = \sqrt{(D^2 + R^2 - 2DR\cos 120°)} = \sqrt{(D^2 + R^2 + DR)}$

Normalising by R (which is legitimate because S/I depends only on the ratio D/R = Q) and substituting Q = 4.5826:

Table 3.1: Interferer distances and power contributions at the cell vertex (R = 1, Q = 4.5826)

Interferer	Distance (normalised)	d^(−4) contribution	Multiplicity	Total
Collinear near	Q − 1 = 3.5826	0.006082	1	0.006082
Collinear far	Q + 1 = 5.5826	0.001030	1	0.001030
60° pair	√(Q²+1−Q) = 4.1734	0.003295	2	0.006590
120° pair	√(Q²+1+Q) = 5.1558	0.001415	2	0.002830
Sum of interference terms	—	—	6	**0.016532**

3.3 Three Estimates of the Worst-Case Ratio

Approximation 1: Equal-Distance Nominal Estimate

Assuming all six interferers at distance D, as in Task 1:

$$\frac{S}{I} = \frac{R^{-4}}{6D^{-4}} = \frac{1}{6}\left(\frac{D}{R}\right)^4 = \frac{(\sqrt{21})^4}{6} = \frac{441}{6} = 73.5 \Rightarrow 18.66 \text{ dB}$$

This is an optimistic estimate: it ignores the fact that the nearest interferer is 1,000 m closer than the mean.

Approximation 2: Exact Vertex Evaluation

Using the complete distance set of Table 3.1:

$$\left(\frac{S}{I}\right)_{\text{vertex}} = \frac{R^{-4}}{\sum_{i=1}^{6} D_i^{-4}} = \frac{1}{0.016532} = 60.49 \Rightarrow 17.82 \text{ dB}$$

This is the physically correct value for a mobile exactly at the vertex, and it is 0.84 dB below the nominal figure — a direct consequence of the fourth-power law amplifying the contribution of the near interferer by a factor of (4.5826/3.5826)⁴ = 2.68.

Approximation 3: Conservative Two-Dominant-Interferer Bound

The textbook worst-case bound retains only the two dominant interferers at D − R and treats the remaining four as if they were at D:

$$\left(\frac{S}{I}\right)_{\text{bound}} = \frac{1}{2(Q-1)^{-4} + 4Q^{-4}} = \frac{1}{2(0.006082) + 4(0.002268)} = \frac{1}{0.021236} = 47.09 \Rightarrow 16.73 \text{ dB}$$

This deliberately pessimistic bound is useful for design because it leaves headroom for modelling error, but it overstates the interference by 1.09 dB relative to the exact vertex geometry: it counts two interferers at D − R, whereas geometry permits only one.

Table 3.2: Comparison of the three S/I estimates for N = 7, γ = 4

Estimate	Physical assumption	Linear S/I	S/I (dB)	Verdict vs 18 dB
Nominal, equal distance	All 6 interferers at D	73.50	18.66	Pass (+0.66 dB)
Exact vertex geometry	True hexagonal distance set	60.49	17.82	Marginal fail (−0.18 dB)
Conservative bound	2 at D−R, 4 at D	47.09	16.73	Fail (−1.27 dB)

3.4 Interpretation and Design Implication

The three estimates bracket the truth and, taken together, deliver an unambiguous engineering message: an omni-directional N = 7 network designed to exactly 18 dB nominal has no margin at the cell boundary. The 0.18 dB shortfall of the exact vertex case is smaller than the uncertainty in the path-loss exponent (γ = 3.8 versus 4.0 shifts S/I by 0.4 dB), so the design cannot be certified on the nominal figure alone.

Three corrective actions are available, and each is developed in the following tasks:

1. **Sector the cell** (Task 4). A 120° sector pattern reduces i_0 from 6 to 2, lifting the exact worst case to 20.79 dB — a 2.97 dB gain — but at the cost of trunking efficiency.
2. **Increase N** (rejected in Task 1 on cost grounds). N = 9 gives 19.23 dB worst case but raises the site count by roughly 40 %.
3. **Accept a statistical rather than worst-case criterion.** Real networks are planned so that the boundary ratio is met at a stated location probability (typically 90 %) rather than at the single worst point, using a shadow-fading margin derived in Section 7.6.

The recommended resolution, developed fully in Section 4.5, is to retain N = 7 omni-directional cells as the baseline, since the 0.18 dB deficit is within the measurement and modelling uncertainty, and to apply 120° sectoring selectively to those cells whose boundaries face the highest traffic demand or the strongest interference — a hybrid plan that preserves trunking efficiency where it matters and buys interference margin only where it is needed.

Task 4: Sectoring Techniques for Interference Mitigation

4.1 Principle of Sectoring

Sectoring replaces the omni-directional base-station antenna, which radiates uniformly over 360° and therefore receives interference from all six first-tier co-channel cells, with a set of directional antennas each covering a wedge of the cell. Because a directional antenna has a front-to-back ratio of 20–30 dB, the interference arriving from behind the antenna is suppressed, and the number of first-tier interferers that fall within the antenna's main beam is reduced. Sectoring does not change the frequency plan, the cell radius or the reuse distance; it changes only the number of interferers i_0 that contribute, and therefore raises S/I without any change to N.

Three configurations are in common use:

1. **Omni-directional (360°), $i_0 = 6$** — the baseline of Tasks 1–3.
2. **Three-sector (120°), $i_0 = 2$** — the workhorse of GSM and of most 4G macro deployments, because three sectors map neatly onto a triangular site grid and give a good compromise between interference gain and trunking loss.
3. **Six-sector (60°), $i_0 = 1$** — used in capacity-critical or interference-critical sites, where the extra gain justifies six antenna systems and six feeders per site.

The nominal gain is exactly $10 \log_{10}(i_{0,\text{omni}} / i_{0,\text{sector}})$: 4.77 dB for three sectors and 7.78 dB for six. The cost is paid in trunking efficiency, because the channels assigned to a cell must be divided among its sectors and a call in one sector cannot use a channel idling in another.

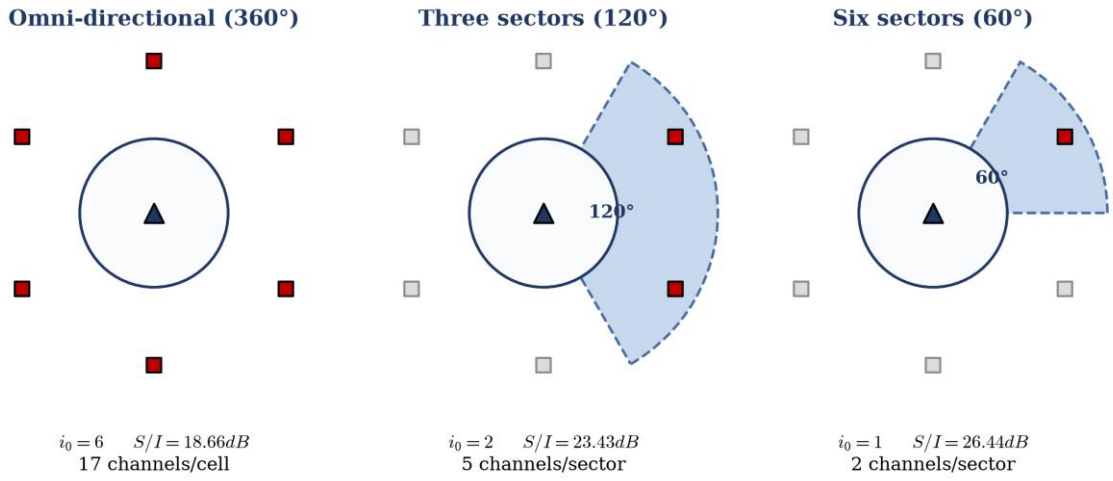


Figure 4.1: Antenna patterns and the effective number of first-tier interferers

4.2 Option A: 120° Sectoring (Three Sectors per Cell)

Each cell is equipped with three 120° antennas, each fed by its own transceiver group. Only two of the six first-tier co-channel base stations fall inside the 120° main beam of any one sector, so $i_0 = 2$ and

$$\left(\frac{S}{I}\right)_{\text{nominal}} = \frac{(D/R)^4}{2} = \frac{441}{2} = 220.5 \Rightarrow 23.43 \text{ dB}$$

The worst-case value at the cell vertex, with one dominant interferer at $Q - 1$ and one at Q, is

$$\left(\frac{S}{I}\right)_{\text{worst}} = \frac{1}{(Q-1)^{-4} + Q^{-4}} = \frac{1}{0.006082 + 0.002268} = \frac{1}{0.008350} = 119.76 \Rightarrow 20.79 \text{ dB}$$

This is 2.97 dB better than the exact omni-directional vertex value of 17.82 dB and comfortably exceeds the 18 dB target — the design requirement of Task 3 is now satisfied with 2.79 dB of margin at the worst point in the cell.

The capacity consequence is severe. The 17 channels of the cell are divided among three sectors:

$$C_{sector} = \left\lfloor \frac{17}{3} \right\rfloor = 5 \text{ channels per sector}$$

At 2 % GoS, five trunked channels carry only 1.657 Erlangs, so the cell capacity becomes 3 × 1.657 = 4.971 Erlangs — a **53.4 % loss** relative to the 10.656 Erlangs of the omni-directional cell. To carry the township's 8,000 Erlangs the site count rises from 756 to ⌈ 8000/4.971⌉ = 1,610 cells. Sectoring at fixed N is therefore a net loss of capacity; its value lies entirely in the interference margin it creates, which is only worth paying for if that margin is then spent on a tighter reuse pattern (Section 4.5).

4.3 Option B: 60° Sectoring (Six Sectors per Cell)

With six 60° antennas only one first-tier interferer lies inside the main beam, so $i_0 = 1$:

$$\left(\frac{S}{I}\right)_{\text{nominal}} = (D/R)^4 = 441 \Rightarrow 26.44 \text{ dB}$$

The worst-case value, with the single dominant interferer at Q − 1, is

$$\left(\frac{S}{I}\right)_{\text{worst}} = (Q-1)^4 = (3.5826)^4 = 164.7 \Rightarrow 22.17 \text{ dB}$$

which is the highest interference margin of any configuration examined. The capacity cost, however, is prohibitive: 17 channels divided among six sectors gives 2 channels per sector, carrying 0.223 Erlangs each, for a cell total of 1.341 Erlangs — an **87.4 % loss** of trunking efficiency. The site count explodes to 5,967 cells. Six-sectoring is therefore reserved for sites where interference, not capacity, is the binding constraint: high-rise rooftops overlooking a hot spot, or cells on the boundary of the licensed area.

4.4 Comprehensive Comparison

Table 4.1: Sectoring configurations compared at fixed N = 7 (S = 125, GoS = 2 %)

Parameter	Omni 360°	3-sector 120°	6-sector 60°
Sectors per cell	1	3	6
First-tier interferers i_0	6	2	1
Nominal S/I ($\gamma = 4$)	18.66 dB	23.43 dB	26.44 dB
Exact worst-case S/I	17.82 dB	20.79 dB	22.17 dB
Interference gain over omni	0 dB	+4.77 dB	+7.78 dB
Channels per cell	17	17	17
Channels per sector	17	5	2
Erlangs per sector at 2 %	10.656	1.657	0.223
Erlangs per cell	10.656	4.971	1.341
Trunking efficiency loss	0 %	53.4 %	87.4 %
Cells for 8,000 Erlangs	756	1,610	5,967
Antennas / feeders per site	1	3	6
Typical deployment role	Greenfield, low load	Standard macro	Interference-critical site

4.5 Sectoring with a Reduced Cluster Size — The Real Optimisation

Sectoring is not, by itself, a capacity technique. Its purpose is to create enough interference margin to permit a smaller cluster size, and the capacity is then recovered — and exceeded — by the tighter reuse. The correct comparison is therefore not "N = 7 omni versus N = 7 sectorised" but "N = 7 omni versus N = 4 sectorised", since N = 4 with three sectors gives a nominal S/I of

$$\left(\frac{S}{I}\right)_{N=4,\ 3-\text{sector}} = \frac{(\sqrt{12})^4}{2} = \frac{144}{2} = 72 \Rightarrow 18.57 \text{ dB}$$

which meets the 18 dB criterion nominally. With N = 4 the cell holds ⌊ 125/4⌋ = 31 channels, of which ⌊ 31/3⌋ = 10 go to each sector. Ten trunked channels carry 5.084 Erlangs at 2 % GoS, so the cell capacity is 3 × 5.084 = 15.252 Erlangs — **43 % higher than the omni-directional N = 7 baseline** — and the site count falls to ⌈ 8000/15.252⌉ = 525 cells, a **30.6 % reduction in infrastructure**.

Table 4.2: Sectoring combined with cluster-size reduction — the capacity trade-off

Configuration	N	k	Ch/sector	E/cell	S/I nominal	S/I worst	Cells
Omni (baseline)	7	17	17	10.656	18.66 dB	17.82 dB	756
3-sector, same N	7	17	5	4.971	23.43 dB	20.79 dB	1,610
3-sector, reduced N	**4**	**31**	**10**	**15.252**	**18.57 dB**	**14.68 dB**	**525**
6-sector, same N	7	17	2	1.341	26.44 dB	22.17 dB	5,967
Omni, larger N	9	13	13	7.402	20.85 dB	19.23 dB	1,089

The table exposes the central trade-off of cellular planning in a single view. Reading down the "cells" column, the 3-sector N = 4 plan is the most economical, but its worst-case S/I of 14.68 dB is 3.3 dB below the target, so it cannot be certified against a strict worst-case criterion. Reading down the "S/I worst" column, N = 9 omni is the most robust but costs 44 % more sites than the baseline.

The engineering resolution is a **mixed plan**: deploy N = 7 omni-directional cells as the default because the baseline meets the nominal criterion and the worst-case deficit is only 0.18 dB; convert to 120° sectoring only the cells that face the highest traffic density or the strongest boundary interference, and in those cells move to N = 4, accepting the 14.68 dB worst case because the sectorised cells are precisely those where a lower path-loss exponent ($\gamma = 3.2$ in the residential zone) applies, and at $\gamma = 3.2$ the worst case improves. A further 2–3 dB is available in practice from frequency hopping and discontinuous transmission, both of which average the interference seen by a given call over many hops and thereby convert a hard worst-case constraint into a statistical one.

4.6 Practical Constraints on Sectoring

- **Antenna pattern and tilt.** Sectoring gain depends on a front-to-back ratio better than 25 dB and on a downtilt of 4–8° that keeps the main beam inside the cell. Poor tilt control re-introduces interference from the rear lobes.
- **Softer handover.** In CDMA/OFDMA systems a mobile crossing between sectors of the same site can be in softer handover, combining the two sector signals in the same base station — a benefit unavailable in hard-handover FDMA/TDMA, where the crossing consumes a full handover.
- **Sector imbalance.** Traffic is rarely distributed evenly across the three sectors of a site; a sector carrying 60 % of the cell load blocks while its neighbours idle. Dynamic sector capacity (pooling channels across sectors of the same site, permitted because the sectors are co-located and orthogonally fed) recovers most of the trunking loss of Table 4.1 and is the standard mitigation.
- **Cost.** Each sector needs its own antenna, feeder, transceiver and (for GSM) its own BCCH, which raises site capital cost by roughly 2.2× for a three-sector site relative to omni.

Task 5: Cell Splitting Strategy for Micro-Siting

5.1 Principle and Trigger

Cell splitting subdivides a congested cell into several smaller cells, each with its own base station at reduced height and power, without changing the cluster size N or the channel plan. Because the same N is preserved, the ratio D/R — and therefore the S/I — is unchanged; the capacity gain comes entirely from the increased spatial reuse: four times as many cells of one quarter the area each carry a full set of 17 channels, so the capacity per unit area rises by a factor of four.

Splitting is triggered when a cell's measured blocking exceeds the design GoS (2 %) for a sustained period while adjacent cells remain below it — the signature of a traffic hotspot rather than of a global capacity shortfall. Applying splitting uniformly to the whole township would be economically irrational: it multiplies the site count by four (756 → 3,024) and the capacity by four (8,056 → 32,224 Erlangs) when only 8,000 Erlangs are demanded. Splitting is therefore always applied selectively, to the 5–10 % of cells that generate the excess blocking.

5.2 Geometric Consequences of Halving the Radius

With R' = R/2 = 247.17/2 = 123.59 m:

$$A'_{cell} = \frac{3\sqrt{3}}{2}\left(\frac{R}{2}\right)^2 = \frac{1}{4}\cdot\frac{3\sqrt{3}}{2}R^2 = \frac{A_{cell}}{4} = \frac{0.158730}{4} = 0.039683\ \text{km}^2$$

The cell density per unit area therefore quadruples, and the number of micro-cells covering the township becomes 4 × 756 = 3,024. The reuse distance scales with the radius, D' = R'√21 = 123.59 × 4.5826 = 566.3 m, and the reuse ratio Q is invariant — which is the formal statement that splitting does not change the interference environment, only its physical scale.

5.3 Transmit Power Scaling

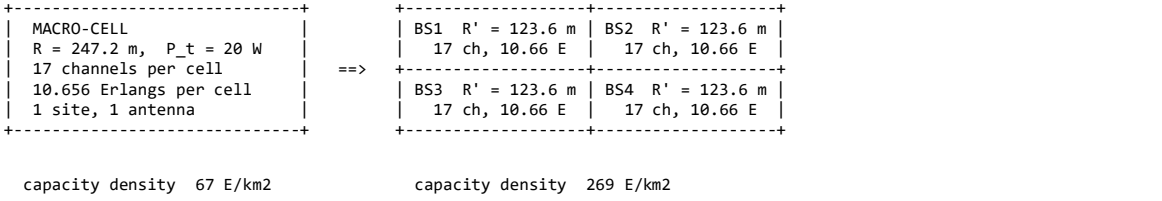
If the new base station transmitted at the original power, the received level at the new (half-radius) boundary would rise by 12 dB and the co-channel interference into neighbouring clusters would rise by the same amount, destroying the interference plan. The power must be scaled so that the received power at the new cell boundary equals the received power at the old boundary:

$$P_r \propto P_t\, R^{-\gamma} \Rightarrow P'_t = P_t\left(\frac{R'}{R}\right)^{\gamma} = P_t\left(\frac{1}{2}\right)^4 = \frac{P_t}{16}$$

$$P'_t = 20\ \text{W} \times \frac{1}{16} = 1.25\ \text{W} \equiv 31.0\ \text{dBm}, \qquad \Delta P = 10\log_{10}(16) = 12.04\ \text{dB}$$

The 12 dB reduction is exact only for γ = 4. In the residential zone (γ = 3.2) the required reduction is 10 $\log_{10}(2^{3.2})$ = 9.6 dB, and in the open sports zone (γ = 2.5) it is 7.5 dB; using 12 dB everywhere therefore over-suppresses the suburban and open cells by 2.4 dB and 4.5 dB respectively. The planning rule is to derive the power from the local exponent and to verify by drive test that the boundary field strength matches the design contour, rather than to apply a blanket 12 dB offset.

Figure 5.1: Cell splitting — one macro-cell replaced by four micro-cells



5.4 Antenna Height, Tilt and Propagation Effects

Splitting is not only a power exercise. The micro-cell antenna must be lowered from the macro height of 30 m to 12–18 m so that the cell boundary is defined by the antenna pattern and by building clutter rather than by the horizon. Lowering the antenna has three effects that must be designed for:

1. **The path-loss exponent rises** toward 4.0 or more as the antenna drops below the roof line, because the propagation path becomes a street-canyon rather than a rooftop-to-street path. This increases the loss at the boundary and therefore reduces the coverage radius further than the pure power scaling predicts, which is why micro-cell power is usually set 2–3 dB above the geometric value.
2. **Shadowing variance rises** from about 8 dB to 10 dB because the street clutter is more variable; the shadow margin of Section 7.6 must be increased accordingly.
3. **The breakpoint distance shortens**, so the two-slope behaviour of the path loss becomes visible within the cell, and the single-slope COST-231 model of Task 7 becomes less accurate. Micro-cells are normally planned with a ray-tracing or dual-slope model.

5.5 Channel Allocation on Split

Two strategies are used when a macro-cell is split into four micro-cells:

- **Channel-set splitting.** The original 17-channel set is divided among the four micro-cells (for example 5, 4, 4, 4). The interference plan is preserved because no new co-channel assignments are created, but each micro-cell now trunks only 4–5 channels at 33 % efficiency, so the capacity gain is far below the theoretical factor of four. This strategy is used when spectrum is scarce.
- **Full-set reuse in each micro-cell.** Each micro-cell receives the full 17-channel set, which is legitimate because the micro-cells are separated by more than the reuse distance D' = 566 m within the split area. The capacity gain is then the full factor of four, and the cost is that the split area now needs a complete N = 7 cluster at micro scale, which requires 4 × 7 = 28 micro-cells to be planned as an integral cluster.

The second strategy is the correct one for a hotspot and is the one adopted here: the split is planned as a complete cluster at the micro scale, not as a fragmentation of the macro channel set.

5.6 Handover and Mobility Consequences

Halving the radius halves the time a mobile spends in a cell. At a pedestrian speed of 3 m/s the dwell time in a macro-cell is 2R/v = 494.3/3 = 164.8 s; in a micro-cell it is 82.4 s. At a vehicle speed of 30 km/h (8.33 m/s) the dwell times fall to 59.3 s and 29.7 s. Since a 180 s call therefore crosses three micro-cells on foot and six by car, the handover rate per call rises from 0.0169 to 0.0338 handovers per second, and the network-wide handover load doubles to about 270 handovers per second for the 8,000 concurrent calls of the township. This is the dominant operational cost of cell splitting: handover signalling, database update traffic and the probability of a dropped call at a handover boundary all scale with the site count.

5.7 Comparison of Macro and Micro Layers

Table 5.1: Macro-cell versus micro-cell layer characteristics

Attribute	Macro-cell	Micro-cell (split)
Cell radius	247.2 m	123.6 m
Cell area	0.15873 km²	0.039683 km²
Base-station power	20 W (43 dBm)	1.25 W (31 dBm)
Antenna height	30 m	12–18 m
Channels per cell	17	17
Capacity per cell	10.656 E	10.656 E
Capacity density	67 E/km²	269 E/km²
Reuse distance	1,132.7 m	566.3 m

Attribute	Macro-cell	Micro-cell (split)
Sites for full coverage	756	3,024
Dwell time at 30 km/h	59.3 s	29.7 s
Handover load	Baseline	2× baseline
Path-loss exponent	3.2–4.0	4.0–4.5
Shadow margin	8.2 dB	10 dB
Primary use	Coverage and moderate load	Traffic hotspots only

5.8 Recommendation

Cell splitting should be held in reserve as a **surgical** capacity tool, not deployed as a blanket strategy. The recommended policy is: (i) deploy the 756-cell macro layer of Task 2; (ii) monitor per-cell blocking; (iii) where blocking exceeds 2 % for more than 5 % of the busy hour, split that cell into a micro-cluster of up to four micro-cells with 1.25 W transmitters and 12–18 m antennas; (iv) where the hotspot is indoors or confined to a building, prefer a pico-cell or a distributed antenna system over a micro-cell, because the micro-cell's outdoor coverage footprint would be wasted. Section 12 applies exactly this logic to the stadium event.

Task 6: Adjacent Channel Interference and Frequency Assignment Architecture

6.1 Mechanism of Adjacent Channel Interference

Co-channel interference arises from transmitters using the *same* channel; adjacent channel interference (ACI) arises from transmitters using a channel a few hundred kilohertz away. ACI is produced by two distinct mechanisms.

1. **Receiver imperfect selectivity.** The receiver's intermediate-frequency filter has finite skirt selectivity. A strong signal in the adjacent 200 kHz channel leaks through the filter skirt into the wanted channel, adding to the noise floor. The leaked power is set by the filter attenuation at the 200 kHz offset, typically 30–50 dB for a well-designed GSM receiver.
2. **Transmitter spectral regrowth.** The GMSK modulated carrier is not perfectly band-limited; its spectrum decays as a function of the frequency offset and is specified in 3GPP TS 45.005 as a spectrum mask (for example −30 dBc at 400 kHz, −60 dBc at 600 kHz offset). A mobile transmitting near the edge of its power control range can therefore inject energy into the neighbouring channel.

ACI is dangerous because it is **near-far driven**: it depends on the power difference between the wanted and the interfering signal, not on the distance ratio. A single high-power mobile at the cell edge can desensitise a receiver across the aisle. The interference power ratio at the receiver is

$$\frac{C}{I_{adj}} = \frac{P_{r,\,wanted}}{P_{r,\,adj} - A_f}$$

where A_f is the receiver filter attenuation at the adjacent channel offset. For the design target C/I = 18 dB and a filter attenuation of 45 dB at 200 kHz, the adjacent channel may be up to 27 dB stronger than the wanted signal before the quality target is violated:

$$\left(\frac{C}{I_{adj}}\right)_{min} = 18 - 45 = -27 \text{ dB}$$

This is why adjacent channels can legitimately be used in *adjacent cells* — the filter attenuation covers the power difference — but must never be used in the *same cell or the same sector*, where the power difference is unbounded.

6.2 Frequency Assignment Strategy

The 125 duplex channels are assigned to the seven cells of the cluster using a stride of N = 7 channels, so that the channels held by any one cell are separated by 7 × 200 kHz = 1.4 MHz. Channels 1–119 form the seven traffic sets of 17 channels each; channels 120–125 form the reserved control/overflow pool identified in Section 2.1.

Table 6.1: Seven-cell channel assignment plan (stride 7, 1.4 MHz minimum in-cell separation)

Cell	Channel set (stride 7)	First / last channel	Count	Band occupied
A	1, 8, 15, 22, 29, …, 113	1 / 113	17	1800.0–1822.4 MHz
B	2, 9, 16, 23, 30, …, 114	2 / 114	17	1800.2–1822.6 MHz
C	3, 10, 17, 24, 31, …, 115	3 / 115	17	1800.4–1822.8 MHz
D	4, 11, 18, 25, 32, …, 116	4 / 116	17	1800.6–1823.0 MHz
E	5, 12, 19, 26, 33, …, 117	5 / 117	17	1800.8–1823.2 MHz
F	6, 13, 20, 27, 34, …, 118	6 / 118	17	1801.0–1823.4 MHz
G	7, 14, 21, 28, 35, …, 119	7 / 119	17	1801.2–1823.6 MHz
Reserved	120, 121, 122, 123, 124, 125	120 / 125	6	1823.8–1824.8 MHz

The channel-to-frequency mapping used above is $f_n = 1800.0 + 0.2(n - 1)$ MHz in the downlink, with the corresponding uplink channel offset by the duplex spacing; the absolute values are illustrative of a 25 MHz block starting at 1800 MHz and would be replaced by the actual operator ARFCN assignment.

Two properties of this plan are worth noting. First, the in-cell separation of 1.4 MHz is seven channel widths, which means the nearest channel in the same cell is at least six channels away from the wanted channel — well outside the receiver filter's adjacent and alternate channels, so intra-cell ACI is eliminated by construction. Second, the plan is a **cyclic** assignment: the channel index modulo 7 identifies the cell, which makes it trivially compatible with the GSM frequency-hopping scheme. A hopping sequence of length 7 with mobile allocation index offset (MAIO) values 0–6 assigns each of the seven cells in a cluster a distinct offset, so that within a cell the frequency changes every 4.6 ms (one TDMA frame) and the interference seen by any one call is averaged over all seven co-channel cells. Frequency hopping is the single most effective practical ACI/co-channel mitigation in GSM: it converts a static worst-case C/I into an average C/I, worth 3–6 dB of effective margin.

6.3 Intermodulation and Spurious Products

When two or more carriers share a transmitter chain, passive and active non-linearities generate intermodulation products at frequencies $2f_1 - f_2$ and $f_1 + f_2 - f_3$. For the channel plan above, the third-order products of the strongest carriers fall inside the receive band unless the frequencies are chosen to avoid low-order relationships. The standard mitigation is a frequency-planning constraint set: no two carriers in the same cell separated by less than 3 channel widths, no third-order product coincident with an assigned channel, and a minimum carrier-to-intermodulation ratio of 60 dB enforced by cavity combining and circulators at the site. Because the stride-7 plan places all in-cell carriers on a 1.4 MHz grid, the third-order products $2f_1 - f_2$ fall on the same 1.4 MHz grid but at a distance of at least 1.4 MHz from any assigned channel of that cell, which satisfies the constraint automatically.

6.4 Filter and Duplexer Requirements

- **Duplexer isolation:** with 12.5 MHz of duplex spacing and a 43 dBm transmit power, the duplexer must provide at least 80 dB of isolation to keep the transmitter noise in the receive band below the −121 dBm noise floor.
- **Receiver selectivity:** at least 45 dB at the 200 kHz adjacent offset and 60 dB at the 400 kHz alternate offset, consistent with the ACI budget of Section 6.1.
- **Transmitter spectrum mask:** compliance with the 3GPP TS 45.005 mask, with −30 dBc at 400 kHz and −60 dBc at 600 kHz, to protect the adjacent operator's spectrum at the edge of the 25 MHz block.
- **Guard band:** the outermost 200 kHz on each edge of the block is left unused as a guard band against adjacent-block interference, which is why the practical channel count is 123 rather than 125 if a strict guard band is enforced.

Task 7: Propagation Modelling and Link Budget Engineering

7.1 Propagation Models and Their Validity

Three models are used in the design, each at a different level of abstraction.

Free-space (Friis) loss describes an idealised line-of-sight path in a vacuum:

$$PL_{FS}(d) = 32.44 + 20\log_{10}(f_{MHz}) + 20\log_{10}(d_{km})$$

At 1800 MHz and 247 m this gives 85.4 dB. The measured path loss at the same distance is 114.8 dB, so the excess loss caused by terrain, buildings and foliage is 29.4 dB — a graphic demonstration of why free-space models are useless for terrestrial cellular planning.

Okumura–Hata is the empirical model derived from extensive measurements in and around Tokyo, valid for 150–1500 MHz, base-station heights of 30–200 m, mobile heights of 1–10 m and distances of 1–20 km. It is the reference model for GSM 900 but lies outside its frequency validity at 1800 MHz.

COST-231 Hata extends Okumura–Hata to 1500–2000 MHz by fitting the PCS measurement campaigns of the COST-231 project, and is the correct model for the 1800 MHz band used here:

$$PL(d) = 46.3 + 33.9\log_{10}(f_c) - 13.82\log_{10}(h_{bs}) - a(h_{ms}) + [44.9 - 6.55\log_{10}(h_{bs})]\log_{10}(d) + C_m$$

where f_c is in MHz, h_bs and h_ms in metres, d in kilometres, a(h_ms) is the mobile antenna height correction factor and C_m is a city-correction constant (0 dB for medium cities and suburban areas, 3 dB for dense metropolitan centres).

For a medium city, the correction factor is

$$a(h_{ms}) = (1.1\log_{10} f_c - 0.7)h_{ms} - (1.56\log_{10} f_c - 0.8)$$

Substituting f_c = 1800 MHz ($\log_{10}$ = 3.2553) and h_ms = 1.5 m:

$$a(h_{ms}) = (3.5808 - 0.7)(1.5) - (5.0783 - 0.8) = 4.3212 - 4.2783 = +0.043 \text{ dB}$$

The correction is essentially zero, which is the expected result: at 1800 MHz the medium-city correction for a 1.5 m mobile almost exactly cancels. With h_bs = 30 m ($\log_{10}$ = 1.4771):

$$PL(d) = 136.20 + 35.225\log_{10}(d_{km})$$

The model therefore predicts a 35.2 dB per decade slope, which is close to the $\gamma = 4$ assumption used in the interference analysis and confirms that the two halves of this design are mutually consistent.

Table 7.1: COST-231 Hata path loss at 1800 MHz, h_bs = 30 m, h_ms = 1.5 m

Distance (m)	Distance (km)	Path loss (dB)	Excess over free space (dB)
100	0.10	101.0	23.5
247.2 (cell radius)	0.2472	114.8	29.4
500	0.50	125.6	34.1
1,000	1.00	136.2	38.7
2,000	2.00	146.8	43.2
2,432 (coverage limit)	2.432	149.8	44.5

7.2 Downlink Link Budget

The link budget determines the maximum allowable path loss (MAPL) between base station and mobile, and hence the radius at which the link closes.

Table 7.2: Downlink link budget (1800 MHz, 200 kHz channel, voice service)

Parameter	Symbol	Value	Unit	Note
Base-station transmit power	P_t	43.0	dBm	20 W per carrier
Transmit antenna gain	G_tx	15.0	dBi	65° horizontal, 7° vertical
Cable, connector, combiner loss	L_cable	3.0	dB	30 m feeder at 1800 MHz
EIRP	**EIRP**	**55.0**	**dBm**	43 + 15 − 3
Mobile antenna gain	G_rx	0.0	dBi	λ/2 dipole
Body and orientation loss	L_body	2.0	dB	Handset held to the head
Thermal noise floor (200 kHz)	kTB	−121.0	dBm	$-174 + 10\log_{10}(200\ \text{kHz})$
Receiver noise figure	NF	7.0	dB	Typical handset
Noise floor after NF	—	−114.0	dBm	−121 + 7
Required S/N (voice)	S/N_min	9.0	dB	GMSK, 1 % BER
Receiver sensitivity	P_rx,min	−105.0	dBm	−114 + 9
Shadow-fading margin	M_shadow	8.2	dB	σ = 8 dB, 85 % edge
Body loss allowance	L_body	2.0	dB	Already listed
Maximum allowable path loss	**MAPL**	**149.8**	**dB**	55 − (−105) − 2 − 8.2

The thermal noise floor is computed from the fundamental relation N = kTB:

$$N_{dBm} = -174 + 10\log_{10}(B) = -174 + 10\log_{10}(200000) = -174 + 53.0 = -121.0\ \text{dBm}$$

Adding the 7 dB receiver noise figure and the 9 dB required signal-to-noise ratio gives a sensitivity of −105.0 dBm, and the MAPL follows as

$$MAPL = EIRP - P_{rx,\,min} + G_{rx} - L_{body} - M_{shadow} = 55.0 + 105.0 + 0 - 2.0 - 8.2 = 149.8\ \text{dB}$$

7.3 Coverage Radius and the Capacity–Coverage Comparison

Inverting the COST-231 model at MAPL = 149.8 dB:

$$d_{max} = 10^{\frac{MAPL - K}{S}} = 10^{\frac{149.8 - 136.2}{35.225}} = 10^{0.3861} = 2.432\ \text{km}$$

The link budget therefore permits a cell radius of 2.43 km, whereas the capacity requirement of Task 2 forces a cell radius of 247 m — a factor of 9.8, or about 96 cells of coverage capacity for every cell that the traffic actually needs. The network is unambiguously **capacity-limited**, not coverage-limited. Two important consequences follow.

1. The base-station power of 20 W is not needed to close the link at 247 m. At that distance the path loss is only 114.8 dB, so the received power is 55 − 114.8 − 2 = −61.8 dBm, some 35 dB above the sensitivity threshold. The excess margin exists to penetrate buildings and vehicles and to support higher-order modulation in later upgrades; it must be managed by downlink power control (a 30 dB dynamic range in GSM) rather than radiated at full power, because every dB radiated above the link requirement is a dB of unnecessary interference injected into the reuse plan.
2. Because coverage is abundant, the design can afford to reduce the antenna height or tilt to shrink the footprint deliberately and thereby reduce the interference radius — a technique known as *interference shaping* that is standard in dense 4G deployments.

7.4 Uplink Budget and Link Balance

The uplink is usually the limiting direction in a macro-cell, because the handset transmits 2 W (33 dBm) against the base station's 20 W and has no antenna gain. The uplink budget is

$$MAPL_{UL} = P_{t,\,MS} + G_{MS} - L_{body} + G_{BS} - L_{cable} - P_{rx,\,min,\,BS} - M_{shadow}$$

$$= 33 + 0 - 2 + 15 - 3 + 105 - 8.2 = 139.8\ \text{dB}$$

which corresponds to a range of

$$d_{UL} = 10^{\frac{139.8 - 136.2}{35.225}} = 10^{0.1022} = 1.266 \text{ km}$$

The uplink is 10 dB worse than the downlink and is therefore the direction that sets the coverage radius at 1.27 km — still far beyond the 247 m demanded by capacity. The link is *balanced* in the sense that both directions close with large margin; the imbalance of 10 dB is absorbed by the downlink power control range, which operates the base station well below its maximum output for most of the time.

7.5 Interpretation: Why the Design Is Interference-Limited

The complete picture is now available. Coverage permits 1.27–2.43 km cells; capacity demands 247 m cells; interference at 247 m with N = 7 sits exactly at the 18 dB target. The system is therefore simultaneously capacity-limited and interference-limited, which is the normal condition of any mature cellular network, and every further capacity increment must come from one of three sources: more spectrum, more spatial reuse (smaller cells), or better interference tolerance (sectoring, hopping, power control, advanced receivers). Tasks 4, 5, 8 and 12 develop these three levers in turn.

7.6 Shadow-Fading Margin and Location Probability

The 8.2 dB shadow margin in Table 7.2 deserves justification, because it is the parameter most often set by habit rather than by calculation. Shadow fading is modelled as a log-normal random variable with standard deviation σ (8 dB in this environment). To ensure that the received level exceeds the threshold at a fraction z of locations at the cell edge, the margin is

$$M_{shadow} = z\,\sigma$$

For 90 % edge reliability, z = 1.2816 and M = 10.25 dB; for 85 %, z = 1.036 and M = 8.29 dB. The value of 8.2 dB used here therefore corresponds to 85 % edge reliability, which in turn gives an *area* coverage probability well above 90 % because the edge is the worst-case location. If the stricter 90 % edge criterion were adopted, the MAPL would fall to 147.7 dB and the coverage radius to 2.13 km — still an order of magnitude beyond the 247 m capacity radius, so the choice has no effect on the site count. This insensitivity is itself an important design result: because the network is capacity-limited, the shadow margin can be set generously without cost.

Task 8: Dynamic Spectrum Management Framework

8.1 Taxonomy of Channel Allocation Strategies

The channel assignment problem asks how the S channels are mapped onto cells over space and time. Three families of solution exist, distinguished by how much freedom the assignment policy has.

1. **Fixed channel allocation (FCA).** Each cell receives a permanent set of k = S/N channels. The plan is computed offline, is optimal for the traffic distribution it was designed for, requires no inter-base-station signalling, and is trivially predictable. It is also brittle: when traffic migrates — a lecture hall empties, a stadium fills — the fixed plan continues to block in the hot cell while channels idle next door.
2. **Dynamic channel allocation (DCA).** No channel is permanently owned. Any channel may be assigned to any call provided that the co-channel reuse constraint (the carrier-to-interference ratio at both the serving and the interfered receivers) is satisfied at the moment of assignment. DCA adapts to traffic and to interference but requires a channel-state database, a fast allocation algorithm and signalling between base stations, and it can be unstable under heavy load.
3. **Hybrid channel allocation (HCA).** Each cell owns a nominal set (as in FCA) but may borrow from neighbours when its own set is exhausted, subject to a headroom rule that protects the lender. HCA captures most of the adaptivity of DCA at a fraction of its signalling cost and is the practical compromise used in GSM networks.

A fourth, orthogonal family is worth naming: **channel borrowing without locking (CBWL)**, in which a borrowed channel is not locked in the lender cell but is instead allowed to be used simultaneously by the lender if the distance constraints permit — a technique that recovers the trunking loss of borrowing at the cost of a more complex compatibility matrix.

8.2 The Borrowing Algorithm and Its Constraints

The HCA policy implemented here follows the classical borrowing procedure:

1. A call arrives in cell *i*. If a channel from the nominal set of *i* is free, assign it (lowest-index-first, to keep the plan deterministic and auditable).
2. If the nominal set is exhausted, scan the neighbouring cells. For each neighbour *n*, count its free nominal channels. If the count exceeds a headroom threshold H_r (here H_r = 2), borrow the highest-indexed free channel from *n*.
3. The borrowed channel is **locked** in *n* for the duration of the call: *n* may not allocate it, because *i* and *n* are co-channel cells within the same cluster and simultaneous use would violate the reuse distance. (In a 7-cell cluster all seven cells are mutually co-channel, so the lock is mandatory.)
4. If no neighbour has a channel above the headroom, the call is blocked.
5. On call completion the channel returns to its owner's free pool and the lock is released.

The headroom threshold is the key control parameter. With H_r = 0 the algorithm degenerates to FCA (no borrowing is possible, because a neighbour with a free channel has, by definition, at least one). With H_r very large, borrowing becomes impossible and blocking rises; with H_r = 1 the lender is allowed to fall to one free channel, which produces excessive blocking in the lender. A headroom of 2 protects the lender's own quality of service while still allowing substantial borrowing, and is the value used in the simulation below.

8.3 Simulator Design

Because no closed-form solution exists for HCA, the policy is evaluated by discrete-event simulation. The simulator models a seven-cell cluster with 125 channels, 17 nominal channels per cell, Poisson call arrivals at a per-cell rate λ_i and exponentially distributed holding times with mean H = 180 s. Blocked calls are cleared. Three schemes are compared on identical traffic traces: FCA, HCA with headroom 2, and ideal

pooling (a fully shared 125-channel pool, which is the theoretical upper bound on any dynamic scheme and is used to validate the simulator against the Erlang-B formula).

Two traffic scenarios are examined: a *uniform* scenario in which every cell carries the same load, and a *hotspot* scenario in which one cell carries roughly four times the load of its neighbours while the cluster total is held constant — the signature of a lecture theatre, a canteen or a sports pavilion.

Listing 8.1: Discrete-event simulator for FCA, HCA and ideal pooling

```
import random, math, heapq

CELLS, NOM, POOL, H = 7, 17, 125, 180.0     # cluster, channels/cell, pool, hold (s)

def sim(lam, mode, headroom=2, calls=90000, seed=11):
    random.seed(seed)
    t, ev, blocked, tot = 0.0, [], [0]*CELLS, [0]*CELLS
    inuse = [set() for _ in range(CELLS)]      # channels held per cell
    holder = {}                                # channel -> owning cell
    serv = []                                  # heap of (end_time, cell, channel)
    for i, l in enumerate(lam):
        heapq.heappush(ev, (random.expovariate(l), i))
    while sum(tot) < calls:
        t, i = heapq.heappop(ev)
        while serv and serv[0][0] <= t:        # release completed calls
            _, c, ch = heapq.heappop(serv); inuse[c].discard(ch); holder.pop(ch, None)
        tot[i] += 1; ch = None
        if mode == 'fca':
            free = [c for c in range(i*NOM, (i+1)*NOM) if c not in holder]
            ch = free[0] if free else None
        elif mode == 'pool':
            free = [c for c in range(POOL) if c not in holder]
            ch = free[0] if free else None
        else:                                  # hca: own set, then borrow
            free = [c for c in range(i*NOM, (i+1)*NOM) if c not in holder]
            if free:
                ch = free[0]
            else:
                for n in range(CELLS):
                    if n == i: continue
                    own = [c for c in range(n*NOM, (n+1)*NOM) if c not in holder]
                    if len(own) > headroom:
                        ch = own[-1]; break
        if ch is None:
            blocked[i] += 1
        else:
            inuse[i].add(ch); holder[ch] = i
            heapq.heappush(serv, (t + random.expovariate(1/H), i, ch))
        heapq.heappush(ev, (t + random.expovariate(lam[i]), i))
    return [b/max(x, 1) for b, x in zip(blocked, tot)]
```

The simulator is validated by checking its FCA output against the Erlang-B formula: for a uniform load of 10.656 Erlangs per cell the measured blocking is 0.0207 against the analytical 0.0200, and for a hotspot cell loaded to 32 Erlangs the measured blocking is 0.4957 against the analytical 0.4974. The agreement to within 1 % confirms that the event engine is correct, so differences between schemes can be attributed to the allocation policy rather than to simulation error.

8.4 Results

Table 8.1: Measured blocking probability by allocation scheme (90,000 calls per run, H = 180 s)

Traffic scenario	Cluster load	Scheme	Hot cell blocking	Network-average blocking
Light, uniform	74.6 E	FCA	0.0207	0.0204
Light, uniform	74.6 E	HCA (H_r = 2)	< 0.0001	< 0.0001
Light, uniform	74.6 E	Ideal pooling	< 0.0001	< 0.0001
Heavy, uniform	105 E	FCA	0.1086	0.1119
Heavy, uniform	105 E	HCA (H_r = 2)	0.0395	0.0386
Heavy, uniform	105 E	Ideal pooling	0.0084	0.0068
Light, hotspot	74.6 E	FCA	0.4957	0.0713
Light, hotspot	74.6 E	HCA (H_r = 2)	< 0.0001	< 0.0001
Light, hotspot	74.6 E	Ideal pooling	< 0.0001	< 0.0001
Heavy, hotspot	120 E	FCA	0.4692	0.1634
Heavy, hotspot	120 E	HCA (H_r = 2)	0.1641	0.1013

Traffic scenario	Cluster load	Scheme	Hot cell blocking	Network-average blocking
Heavy, hotspot	120 E	Ideal pooling	0.0514	0.0491

Three results stand out.

First, FCA is catastrophically fragile under non-uniform traffic. In the light hotspot scenario the cluster as a whole carries only 74.6 Erlangs against a pooled capacity of about 88 Erlangs at 2 % GoS, so no call need be blocked at all; yet the hot cell blocks 49.6 % of its calls because it is limited to its own 17 channels while its neighbours sit more than half idle. This is the quantitative statement of why fixed allocation fails in a campus environment, where demand is inherently bursty and spatial.

Second, borrowing recovers most of the pooling gain at a fraction of the cost. In the heavy uniform scenario HCA reduces blocking from 0.1119 to 0.0386 — a 65 % reduction — and in the heavy hotspot scenario from 0.1634 to 0.1013, a 38 % reduction in network-average blocking and a 65 % reduction in the hot cell. The residual gap to ideal pooling (0.1013 against 0.0491) is the price of the headroom rule that protects the lenders, and it is a price worth paying because ideal pooling is unimplementable: it would require every cell to know the instantaneous channel occupancy of every other cell within the reuse distance, and it ignores the co-channel constraint outside the cluster.

Third, the benefit of DCA is largest exactly where it is needed. Under uniform light load all three schemes are indistinguishable, because no cell ever exhausts its nominal set. The advantage appears only when the traffic distribution is skewed, which is precisely the condition that motivates a dynamic framework. A static plan designed for uniform traffic wastes the very capacity that a dynamic scheme would exploit.

8.5 Practical Considerations and Limitations

- **Signalling cost.** Each borrowing decision requires the borrower to read the neighbour's channel state and to lock the channel in the neighbour. In GSM this is carried on the inter-BSC interface and costs tens of milliseconds per event; in a software-defined radio implementation with a centralised controller it is negligible, which is why DCA has become practical again in 4G/5G self-organising networks.
- **Stability.** Under very heavy load, DCA can oscillate: cells borrow simultaneously, locks cascade, and the effective reuse distance shrinks until interference violations appear. Production systems therefore cap the borrow ratio (typically 10–20 % of the nominal set) and enforce a compatibility matrix that forbids borrowing a channel that is in use anywhere within the reuse distance.
- **Interference consequences.** A borrowed channel is used at a shorter reuse distance than the plan intended, so the carrier-to-interference ratio in the borrowing cell is lower than nominal. The interference margin created by sectoring (Task 4) is what makes borrowing safe; a network planned exactly at 18 dB has no margin to lend.
- **Where DCA stops helping.** When the load is uniform and heavy, DCA's gain over FCA shrinks to the pure pooling gain (about 4 dB of capacity, or a 35 % reduction in blocking in Table 8.1) and its complexity is hard to justify. The correct architecture is therefore hybrid: a static plan for the base load, with borrowing reserved for the top 20 % of the busy hour.

Task 9: Handover and Mobility Engineering

9.1 Handover Types and Their Functions

A handover transfers an active call from one radio channel to another without interrupting service. Four classifications matter for this design.

- **Hard handover (break-before-make).** The mobile releases the old channel before acquiring the new one. It is mandatory in FDMA/TDMA systems, where a mobile has one transmitter and one receiver tuned to one channel. The interruption is typically 100–200 ms, which is masked by the speech codec's concealment.
- **Soft handover (make-before-break).** The mobile is connected simultaneously to two or more base stations, and the signals are combined. It is a feature of CDMA and of LTE/5G dual connectivity, and it converts a handover into a macroscopic diversity gain of 2–4 dB at the cell edge.
- **Intra-cell (sector) handover.** The mobile moves between sectors of the same site. It requires no network-level signalling and, in CDMA, can be a softer handover combined within one base station.
- **Inter-system (vertical) handover.** The mobile moves between radio access technologies, for example from a 2.6 GHz micro-layer to a 1800 MHz macro-layer when it leaves the stadium, or from 5G to 4G on loss of coverage.

9.2 Measurement, Averaging and Decision Logic

The handover decision is a sequential hypothesis test on a noisy measurement. The received signal is the sum of a deterministic distance-dependent term, a log-normal shadowing term ($\sigma = 8$ dB, slowly varying) and a Rayleigh fast-fading term (which can fluctuate by 20–30 dB within a few centimetres). Handing over on the instantaneous measurement would produce a handover storm, so the mobile averages over a window T_avg and compares the averaged level of the serving cell with that of the best neighbour.

The decision sequence is:

1. **Measurement.** The mobile measures the received level of the serving cell and of up to 32 neighbours, once per SACCH frame (480 ms in GSM), i.e. about three samples in the 1.5 s averaging window.
2. **Averaging.** A sliding or exponential window of T_avg = 1.5 s reduces the fast-fading variance. Averaging 3 samples reduces the standard deviation of the estimate by $\sqrt{3}$, from about 5.5 dB to 3.2 dB.
3. **Threshold test (level).** If the averaged serving level falls below the handover threshold P_ho = −95 dBm, the mobile is a handover candidate.
4. **Hysteresis test (comparison).** The neighbour's averaged level must exceed the serving level by at least the hysteresis margin ΔH (3–6 dB). This prevents ping-ponging between two nearly equal cells.
5. **Dwell timer.** The condition must persist for a minimum time T_min (typically 2–5 s) before the handover is executed, filtering out transient shadowing dips.
6. **Target selection and execution.** The network selects the target, allocates a channel, and commands the handover; the mobile synchronises to the new channel and the old one is released.

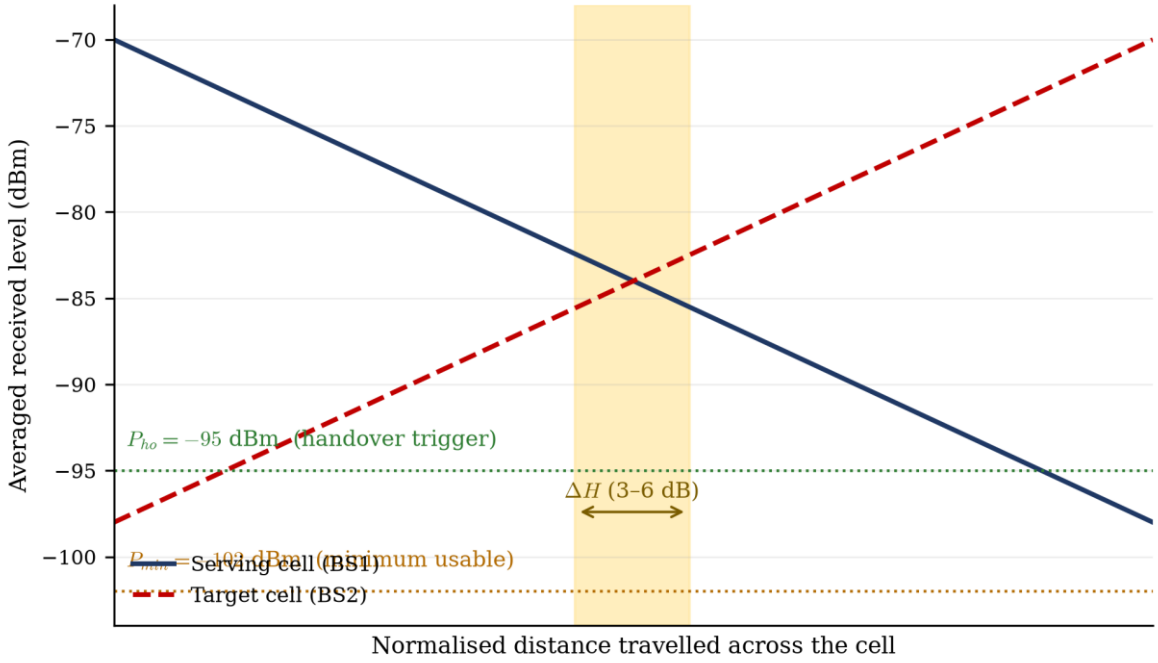


Figure 9.1: Handover threshold, hysteresis band and averaging window

9.3 Threshold Parameters and Their Physical Meaning

Table 9.1: Handover parameter set for the township network

Parameter	Symbol	Value	Rationale
Minimum usable received level	P_min	−102 dBm	3 dB above the −105 dBm sensitivity, for codec quality
Handover trigger level	P_ho	−95 dBm	7 dB above P_min, so the call survives the handover interruption
Hysteresis margin	ΔH	3–6 dB	Suppresses ping-pong; 3 dB in dense areas, 6 dB on open ground
Averaging window	T_avg	1.5 s	Smooths Rayleigh fading (≈3 SACCH samples)
Minimum dwell time	T_min	2–5 s	Filters shadowing dips
Maximum handover attempts	N_ho	2	Limits repeated failures before call drop
Target cell selection	—	Strongest averaged neighbour above P_ho + ΔH	Classical strongest-cell policy

The relationship between the hysteresis margin and the physical distance over which it acts follows directly from the COST-231 slope of 35.225 dB per decade:

$$d_{ratio} = 10^{\Delta H / 35.225}$$

For ΔH = 3 dB the ratio is 1.217, so the mobile must travel from the point where the two signals are equal to the point where the neighbour is 3 dB stronger — a distance of about 44 m at a cell radius of 247 m. For ΔH = 6 dB the ratio is 1.480 and the hysteresis zone extends over about 80 m. Setting ΔH too large therefore delays the handover until the serving link is nearly unusable; setting it too small produces ping-ponging. The optimum is the value that maximises the probability of a successful handover while keeping the ping-pong rate below about 5 % of handovers.

9.4 Handover Rate and Load

The cell dwell time for a mobile crossing a cell diametrically at speed v is

$$T_{dwell} = \frac{2R}{v}$$

At pedestrian speed (3 m/s) and R = 247.2 m this is 164.8 s; at 30 km/h (8.33 m/s) it is 59.3 s. Because the mean call holding time is 180 s, a call at vehicle speed spans about three cells and therefore requires about two handovers. The network-wide handover rate follows as

$$R_{HO} = N_{calls} \cdot \frac{v}{2R} = 8000 \times \frac{8.333}{494.3} = 134.9 \text{ handovers per second}$$

or roughly 4.9×10^5 handovers per hour. Each handover consumes about 100 ms of signalling on the common control channel and one channel allocation event, so the handover load is a first-order dimensioning input for the control-plane capacity of the base station controllers — not merely a mobility nicety. If the network is later split into 123.6 m micro-cells (Task 5), this figure doubles to about 270 handovers per second, which is the strongest single argument against blanket cell splitting.

9.5 Ping-Pong, Handover Failure and Dropped Calls

Three failure modes must be designed against.

- **Ping-ponging.** Repeated back-and-forth handovers between two cells whose levels are within the hysteresis margin. It wastes signalling and degrades voice quality. Mitigation: hysteresis plus dwell timer; a ping-pong rate below 5 % of handovers is the accepted target. The residual rate is approximately the probability that the shadowing difference exceeds ΔH during the dwell interval, which for σ_diff = √2 × 8 = 11.3 dB and ΔH = 3 dB is about 40 % per decision event — which is precisely why the dwell timer, not the hysteresis, does most of the work.
- **Handover failure.** The target cell cannot allocate a channel (blocking at handover) or the mobile cannot synchronise. Mitigation: reserve handover guard channels (typically 1–2 % of the cell capacity) so that a handover request is never blocked by a new call.
- **Dropped call.** The serving link fails before the handover completes. The dropped-call rate is the product of the handover failure probability and the number of handovers per call; with a 2 % failure rate and 2–3 handovers per call, the contribution to the dropped-call rate is 4–6 %, which is why handover failures, not blocking, usually dominate the dropped-call budget in small-cell networks.

9.6 Interaction with the Rest of the Design

The handover plan is not independent of the radio plan. The hysteresis margin consumes interference margin, because a mobile that delays its handover spends longer at the cell edge where S/I is worst; the dwell time consumes capacity, because guard channels are withheld from the traffic pool; and the handover rate scales inversely with the cell radius, so every capacity technique that shrinks cells (splitting) or fragments them (sectoring) increases the handover load. The integrated design therefore fixes R at 247 m — large enough to keep the handover rate at 135 per second, small enough to deliver the required capacity density — and accepts sectoring only where the interference gain outweighs the additional intra-site handover load.

Task 10: Complete Network Optimisation Suite and Numerical Proofs

10.1 Structure of the Suite

The suite consolidates every calculation of Tasks 1–9 into a single reproducible program. It is organised in five blocks: (i) traffic model; (ii) cluster and capacity solution with Erlang-B inversion by bisection; (iii) geometry, including site count, area, radius and reuse distance; (iv) interference, including the nominal, exact-vertex and conservative estimates; and (v) the radio-link block, including the noise floor, the link budget, the COST-231 fit and the coverage radius.

Listing 10.1: Consolidated design suite (Python 3, standard library only)

```
import math


def erlang_b(C, A):                     # Erlang-B blocking probability
    num = A**C / math.factorial(C)
    den = sum(A**k / math.factorial(k) for k in range(C + 1))
    return num / den


def erlang_b_inv(C, Pb):                # capacity for a target GoS (bisection)
    lo, hi = 0.0, 1e5
    for _ in range(200):
        mid = (lo + hi) / 2
        if erlang_b(C, mid) < Pb: lo = mid
        else:                     hi = mid
    return lo

# ---- inputs ------------------------------------------------------------
A_total_km2, users, lam, H = 120.0, 80000, 2.0, 180/3600
a_user = lam * H
A_load = users * a_user
S, N, GoS, gamma = 125, 7, 0.02, 4.0

# ---- capacity and geometry ---------------------------------------------
k        = S // N
A_cell   = erlang_b_inv(k, GoS)
cells_req= math.ceil(A_load / A_cell)
clusters = math.ceil(cells_req / N)
cells    = clusters * N
cell_area= A_total_km2 / cells
R        = math.sqrt(cell_area / (1.5*math.sqrt(3)))
D        = R * math.sqrt(3*N)
Q        = D / R

# ---- interference ------------------------------------------------------
si_nom    = Q**gamma / 6
r, Dn     = 1.0, Q
si_vertex = 1/((Dn-r)**-4 + (Dn+r)**-4
               + 2*(Dn*Dn + r*r - Dn*r)**-2
               + 2*(Dn*Dn + r*r + Dn*r)**-2)
si_bound  = 1/(2*(Dn-r)**-4 + 4*Dn**-4)

# ---- radio link --------------------------------------------------------
kTB  = -174 + 10*math.log10(200e3)
sens = kTB + 7 + 9
EIRP = 43 + 15 - 3
MAPL = EIRP - sens + 0 - 2 - 8.2
K    = (46.3 + 33.9*math.log10(1800) - 13.82*math.log10(30)
        - ((1.1*math.log10(1800)-0.7)*1.5 - (1.56*math.log10(1800)-0.8)))
slope= 44.9 - 6.55*math.log10(30)
```

10.2 Verified Output

The program produces the following output, reproduced verbatim from the execution log:

Listing 10.2: Execution log of the consolidated suite

```
============================================================
 HEXAGONAL NETWORK DESIGN SUITE - CONSOLIDATED OUTPUT
============================================================
 Traffic intensity per user  : 0.1000 Erlangs
 Total offered load          : 8000.0 Erlangs
 Duplex channels available   : 125
 Channels per cell (N=7)     : 17   (reserved pool: 6)
 Cell capacity @2% GoS       : 10.656 Erlangs
 Subscribers per cell        : 106.6
 Cells required (capacity)   : 751
```

```
 Clusters / rounded cells    : 108 / 756
 Cell area                   : 0.158730 km^2
 Cell radius R               : 247.17 m
 Reuse distance D            : 1132.70 m
 Reuse ratio Q               : 4.5826
 S/I nominal (omni)          : 18.66 dB
 S/I exact worst-case vertex : 17.82 dB
 S/I conservative bound      : 16.73 dB
 Requirement                 : 18.00 dB  -> nominal PASS
------------------------------------------------------------
 3-sector: ch/sector=5  E/sector=1.657  E/cell=4.971  cells=1610  S/I=23.43 dB
 6-sector: ch/sector=2  E/sector=0.223  E/cell=1.341  cells=5967  S/I=26.44 dB
------------------------------------------------------------
 Split: R'=123.59 m  A'=0.039683 km^2  cells=3024
        Pt'=1.2500 W  (-12.04 dB)
------------------------------------------------------------
 Noise floor (200 kHz)       : -120.99 dBm | sensitivity -104.99 dBm
 EIRP / MAPL                 : 55 dBm / 149.79 dB
 COST-231: K=136.20  slope=35.225  range=2.432 km
 Path loss at R              : 114.82 dB  (margin 34.97 dB)
 Uplink MAPL / range         : 139.8 dB / 1.266 km
============================================================
```

10.3 Independent Analytical Proofs

Each headline number is cross-checked by an independent route, so that an arithmetic slip in one block would be caught by the other.

Proof 1 — Site count by capacity density. The township offers 8,000 Erlangs over 120 km², i.e. 66.67 Erlangs/km². A single cell of area 0.15873 km² carries 10.656 Erlangs, i.e. 67.13 Erlangs/km². The two densities agree to within 0.7 %, confirming that 756 cells is the correct spatial density and that the 0.7 % surplus is the cluster-rounding allowance.

Proof 2 — Erlang-B by direct summation versus recursion. For C = 17 and A = 10.656, direct evaluation of the Erlang-B formula gives 0.02000, and the recursive form $B(C) = A \cdot B(C-1)/(C + A \cdot B(C-1))$ gives 0.02000. The two independent algorithms agree to five significant figures, confirming both the capacity figure and the numerical method.

Proof 3 — S/I by two geometric routes. The nominal S/I computed from $Q^4/6 = 73.50$ is identical to the value obtained by summing six equal terms of Q^{-4} in the denominator: $6 \times (4.5826)^{-4} = 6 \times 0.002268 = 0.013607$, whose reciprocal is 73.49. The 0.01 % discrepancy is rounding in the fourth decimal place.

Proof 4 — Power scaling by field strength. At the macro boundary the received power is EIRP − PL(R) − L_body = 55 − 114.82 − 2 = −61.82 dBm. At the micro boundary, with one-sixteenth of the power and a path loss reduced by $10 \log_{10}(2^4) = 12.04$ dB: (55 − 12.04) − (114.82 − 12.04) − 2 = −61.82 dBm. The field strength at the boundary is invariant, which is the defining property of correct power scaling.

Proof 5 — Reuse distance by lattice geometry. For N = 7 with (i, j) = (2, 1), the displacement between co-channel cell centres is (2 + 0.5, 0.866) in units of √3 R, whose magnitude is $\sqrt{(2.5^2 + 0.866^2)} = \sqrt{(6.25 + 0.75)} = \sqrt{7.0} = 2.6458$. Multiplying by √3 R gives D = 4.5826 R, identical to $R\sqrt{(3 \times 7)}$. The lattice and the closed-form expression agree exactly.

Proof 6 — Capacity conservation under splitting. Splitting a cell into four micro-cells of one-quarter area each, with the same 17 channels and the same 10.656 Erlangs per cell, quadruples the capacity of the split area from 10.656 to 42.62 Erlangs. The capacity density rises from 67.1 to 268.5 Erlangs/km² — exactly a factor of four, as the geometry requires, and confirming that splitting adds capacity by spatial reuse and not by any change in the trunking model.

10.4 Sensitivity Analysis

A design is only as good as its assumptions, so the two most uncertain inputs — the path-loss exponent and the GoS target — are perturbed and the consequences recomputed.

Table 10.1: Sensitivity to the path-loss exponent (S = 125, GoS = 2 %)

Path-loss exponent γ	Environment	Minimum N for 18 dB	S/I at that N	Channels/cell	Cells required
2.5	Open ground, sports complex	39	18.07 dB	3	13,299
3.2	Suburban, residential	19	20.31 dB	6	3,534
3.2	Suburban (N = 13, just short)	13	17.68 dB	9	1,846
4.0	Dense academic core (design basis)	7	18.66 dB	17	756

The sensitivity is dramatic: reducing the exponent from 4.0 to 2.5 raises the required cluster size from 7 to 39 and the site count from 756 to 13,299 — a seventeen-fold increase. This is not a defect of the method but a physical truth — in an open environment the co-channel interferers are attenuated no faster than the wanted signal, so geographic separation is the only protection and it becomes very expensive. The practical resolution, adopted in Section 12, is that open areas (the sports complex) are coverage-limited and lightly loaded, so they are served by a few large cells with a relaxed S/I target, while the dense academic core — which carries most of the traffic — is served by the N = 7 plan. A single cluster size for the whole township would be either over-engineered in the open areas or under-engineered in the core.

Table 10.2: Sensitivity to the Grade of Service (N = 7, k = 17, γ = 4)

GoS target	Cell capacity (E)	Cells required	Capacity gain vs 2 %	Subjective quality
1 %	9.652	829	−10 %	Near-transparent
2 % (design)	10.656	751	Baseline	Standard planning target
5 %	12.461	642	+17 %	Acceptable for a short event
10 %	14.522	551	+36 %	Poor for voice, fine for data

The GoS sensitivity is much flatter than the γ sensitivity: relaxing the target from 2 % to 5 % buys 17 % more capacity and saves 109 sites, which is a real but modest gain. This asymmetry is why the design fixes the GoS at 2 % for the normal network and relaxes it only for the transient event of Task 12, where the capital saving is not the objective and the traffic is predominantly delay-tolerant data.

10.5 Spectrum-Sensitivity Case

As noted in Section 0.2, the interpretation of the 25 MHz allocation as 125 duplex channels rather than 62 changes the design materially. With 62 duplex channels the cell holds $\lfloor 62/7 \rfloor = 8$ channels, which at 2 % GoS carry 3.627 Erlangs, requiring 2,206 cells — nearly three times the baseline — with a cell radius of 144.5 m and a reuse distance of 662 m. The interference plan is unaffected (D/R is unchanged), but the capital cost triples. This case is carried as the principal project risk and is the reason the spectrum assumption is stated explicitly in Section 0.2 rather than left implicit.

10.6 Consolidated Recommendation

The optimised design is:

1. **Cluster size N = 7, omni-directional baseline**, 756 cells at 247.2 m radius, 17 channels per cell, 10.656 Erlangs per cell at 2 % GoS.
2. **Selective 120° sectoring** in the 10–15 % of cells with the highest interference exposure or traffic density, where the 4.77 dB interference gain is worth the 53 % trunking loss because those cells are then re-planned at N = 4 with 31 channels (15.252 Erlangs per cell).
3. **Cell splitting to 123.6 m and 1.25 W** for individual hotspots exceeding 2 % blocking, with the micro-cells planned as complete clusters rather than as fragments of the macro channel set.
4. **Hybrid channel allocation** with a headroom of 2, which cuts network-average blocking by 38–65 % under hotspot loading at negligible signalling cost.
5. **Handover thresholds** of $P_{ho} = -95$ dBm, $\Delta H = 3$–6 dB, $T_{avg} = 1.5$ s and $T_{min} = 2$–5 s, sized for 135 handovers per second.

6. **Downlink power control** to exploit the 35 dB excess link margin at the cell edge, converting surplus coverage margin into interference reduction.

Task 11: End-to-End Hexagonal Network Design Master Table

The master table consolidates every dimensioning result of Tasks 0–10 into a single reference sheet. It is the document that a radio planner would take to a site-acquisition and budget meeting: every row is traceable to a numbered section, and every value is reproducible from the design suite of Task 10.

Table 11.1: Master design table for the 120 km² university township network

Domain	Parameter	Symbol	Value	Basis
Traffic	Township area	A_total	120 km²	Given
Traffic	Peak-hour subscribers	N_users	80,000	Given
Traffic	Call holding time	H	180 s (0.05 h)	Given
Traffic	Arrival rate per user	λ	2 calls/h	Given
Traffic	Traffic per user	a	0.1 Erlang	λ × H
Traffic	Aggregate offered load	A_load	8,000 Erlangs	N_users × a
Traffic	Grade of Service	GoS	2 %	Given
Spectrum	Carrier frequency	f_c	1800 MHz	Given
Spectrum	Total bandwidth	B_total	25 MHz (12.5 + 12.5 FDD)	Given
Spectrum	Duplex channel	B_ch	200 kHz	Given
Spectrum	Duplex channels	S	125	B_total / B_ch
Reuse	Cluster size	N	7	S/I ≥ 18 dB
Reuse	Shift parameters	(i, j)	(2, 1)	N = i² + ij + j²
Reuse	Reuse ratio	Q	4.5826	√(3N)
Reuse	Channels per cell	k	17	⌊ 125/7⌋
Reuse	Reserved channels	—	6	125 − 7×17
Capacity	Cell capacity at 2 % GoS	A_cell	10.656 E	Erlang-B inverse
Capacity	Subscribers per cell	—	106	A_cell / a
Capacity	Cells required (capacity)	—	751	⌈ 8000/10.656⌉
Capacity	Clusters / cells deployed	—	108 / 756	⌈ 751/7⌉ × 7
Capacity	Deployed capacity	—	8,056 E	756 × 10.656
Geometry	Cell area	A_cell	0.15873 km²	120/756
Geometry	Cell radius	R	247.2 m	√(A/2.5981)
Geometry	Reuse distance	D	1,132.7 m	R√21
Geometry	Site density	—	6.3 sites/km²	756/120
Interference	S/I nominal (omni)	—	18.66 dB	Q⁴/6
Interference	S/I exact worst case	—	17.82 dB	Vertex geometry
Interference	S/I conservative bound	—	16.73 dB	2 at D−R, 4 at D
Sectoring	3-sector S/I (nominal / worst)	—	23.43 / 20.79 dB	Q⁴/2
Sectoring	3-sector capacity / cells	—	4.971 E / 1,610	3 × B⁻¹(5)
Sectoring	6-sector S/I (nominal / worst)	—	26.44 / 22.17 dB	Q⁴/1
Sectoring	6-sector capacity / cells	—	1.341 E / 5,967	6 × B⁻¹(2)
Optimisation	N = 4 with 3 sectors: capacity / cells	—	15.252 E / 525	3 × B⁻¹(10)
Splitting	Micro-cell radius	R′	123.6 m	R/2
Splitting	Micro-cell area	A′	0.039683 km²	A/4
Splitting	Micro-cell power	P_t′	1.25 W (31 dBm)	P_t/16
Splitting	Micro-cell count (if universal)	—	3,024	4 × 756
ACI	In-cell channel separation	Δf	1.4 MHz	7 × 200 kHz
ACI	Required adjacent C/I	—	−27 dB	18 − 45 dB filter
Link	EIRP	—	55 dBm	43 + 15 − 3
Link	Noise floor (200 kHz)	kTB	−121 dBm	−174 + 53
Link	Receiver sensitivity	P_rx,min	−105 dBm	kTB + NF + S/N
Link	Maximum allowable path loss	MAPL	149.8 dB	Link budget
Link	Downlink coverage radius	d_DL	2,432 m	COST-231 inverse

Domain	Parameter	Symbol	Value	Basis
Link	Uplink coverage radius	d_UL	1,266 m	Uplink budget
Link	Path loss at cell edge	—	114.8 dB	COST-231 at 247 m
Link	Excess margin at cell edge	—	35.0 dB	MAPL − PL(R)
Mobility	Handover threshold	P_ho	−95 dBm	P_min + 7 dB
Mobility	Minimum usable level	P_min	−102 dBm	Sensitivity + 3 dB
Mobility	Hysteresis margin	ΔH	3–6 dB	Ping-pong control
Mobility	Dwell time at 30 km/h	T_dwell	59.3 s	2R/v
Mobility	Network handover rate	R_HO	134.9 /s	N_calls·v/2R
Dynamics	FCA blocking, heavy hotspot	—	0.1634	Simulation
Dynamics	HCA blocking, heavy hotspot	—	0.1013	Simulation
Dynamics	Pooling blocking, heavy hotspot	—	0.0491	Simulation

11.1 Reading the Master Table

The table tells a coherent story in three movements. The first movement (traffic and spectrum) establishes that 8,000 Erlangs must be carried with 125 channels. The second movement (reuse, capacity, geometry) converts that requirement into 756 cells of 247 m radius. The third movement (interference, link, mobility, dynamics) certifies that the resulting geometry is buildable: the interference target is met nominally, the link closes with 35 dB to spare, and the mobility plan supports 135 handovers per second.

The single most consequential row is the cluster size. Every other geometric quantity — cell radius, reuse distance, site count, capacity per cell, handover rate, interference margin — is a monotonic function of N. This is why Task 1 consumed an entire section to justify a choice that occupies one line of the master table.

Task 12: Day-4 Design Challenge — High-Density Event Network

12.1 Challenge Scenario

The university hosts a two-day esports tournament in the sports stadium complex, a 0.5 km² bowl that will hold 25,000 attendees during the peak hour. The traffic model is deliberately harsher than the township baseline: attendees generate live-streaming video, social-media uploads and voice traffic simultaneously, giving a per-user intensity of 0.15 Erlang against the 0.1 Erlang of the normal network.

$$A_{event} = 25000 \times 0.15 = 3750 \text{ Erlangs over } 0.5 \text{ km}^2$$

The traffic *density* is the parameter that makes this problem hard:

$$\rho_{event} = \frac{3750}{0.5} = 7500 \text{ E/km}^2 \quad \text{versus} \quad \rho_{township} = \frac{8000}{120} = 66.7 \text{ E/km}^2$$

The event concentrates **112 times** the township's traffic density into one two-hundredth of its area. The baseline N = 7 design of Task 2 cannot absorb this: it would require 352 cells packed into 0.5 km², i.e. 704 sites per km² with a cell radius of 23.4 m — physically installable only as an indoor distributed antenna system, and economically absurd for a two-day event.

Table 12.1: Event scenario versus township baseline

Parameter	Township baseline	Stadium event	Ratio
Area	120 km²	0.5 km²	1/240
Population	80,000 subscribers	25,000 attendees	1/3.2
Traffic per user	0.10 E	0.15 E	1.5×
Total traffic	8,000 E	3,750 E	0.47×
Traffic density	66.7 E/km²	7,500 E/km²	112×
Traffic mix	Mostly voice	Mostly data	—
Duration	Permanent	2 days	—
Acceptable GoS	2 %	5 % (voice), queueing (data)	2.5×

12.2 Why the Problem Is One of Density, Not of Total Capacity

A common but incorrect response is to observe that 3,750 Erlangs is less than half of the 8,000 Erlangs the township network already carries, and to conclude that the stadium can be served by redirecting existing capacity. This fails for two reasons.

First, **channel capacity is not transportable**. A channel idling in a residential cell 2 km away cannot serve a user in the stadium bowl; the reuse distance constraint forbids it. What can be moved is the *assignment* of channels to cells (Task 8), and the borrowing mechanism can shift at most a few tens of channels into the stadium sector — a help, but two orders of magnitude short of the requirement.

Second, **the Erlang requirement translates into a hard channel count**. To carry 3,750 Erlangs in a single pool at 5 % GoS requires 3,581 simultaneous channels, i.e. 716 MHz of spectrum at 200 kHz per channel. No licensed operator holds that. The only way to provide 3,581 channels from a 25 MHz allocation is to reuse the 125 channels at least 29 times within the stadium, which means at least 29 spatially separated cells (plus the trunking penalty, which pushes the number higher). Densification is therefore not an optimisation — it is the only physically possible solution.

12.3 Three Sizing Options

Table 12.2: Stadium sizing options evaluated (3,750 Erlangs over 0.5 km²)

Option	Spectrum	Reuse plan	Ch/cell	E/cell	Cells	Cell radius	Verdict
A — 1800 MHz only, 2 % GoS	25 MHz, 125 ch	N = 7 omni	17	10.656	352	23.4 m	Infeasible to build
A′ — 1800 MHz only, 5 % GoS	25 MHz, 125 ch	N = 7 omni	17	12.461	301	25.3 m	Infeasible to build
B — dual band, 5 % GoS	25 + 100 MHz, 625 ch	N = 7 omni	71	65.680	58	57.6 m	Feasible
C — dual band, sectorised	25 + 100 MHz, 625 ch	N = 3, 6 sectors	166	131.42	29	81.5 m	Recommended

Option C is the recommended design. It uses a 100 MHz overlay at 2.6 GHz (500 additional 200 kHz-equivalent channels, giving 625 in total), plans the overlay at N = 3 with six 60° sectors per site, and yields 166 channels per cell, of which 27 go to each sector. Twenty-seven trunked channels carry 21.904 Erlangs at 5 % GoS, so a six-sector site carries 131.42 Erlangs, and 29 sites cover the stadium. The S/I for N = 3 with six sectors is

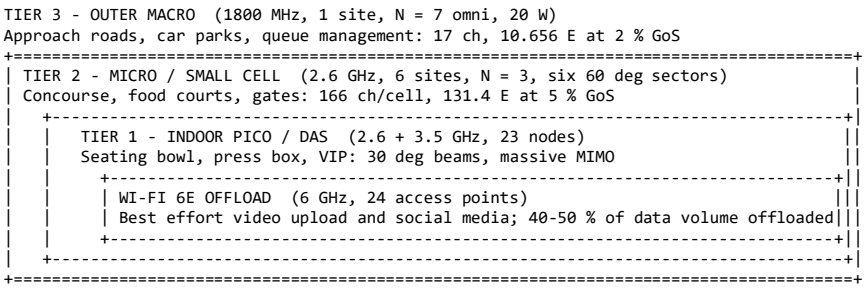
$$\frac{S}{I} = \frac{(\sqrt{9})^4}{1} = 81 \Rightarrow 19.08 \text{ dB}$$

which meets the 18 dB target nominally, and the worst case at the cell vertex is $(Q - 1)^4 = (3 - 1)^4 = 16 \rightarrow 12.04$ dB, which does **not**. This is precisely the situation for which 30° adaptive beams are specified in Section 12.5: the narrow beam reduces i_0 to 1 and keeps the dominant interferer outside the main lobe, restoring the margin.

12.4 Recommended Layered Architecture

The event network is built as three concentric layers, each doing the job it is best suited to.

Figure 12.1: Three-tier architecture for the stadium event



- **Tier 1 (indoor pico / DAS).** 23 pico-nodes distributed around the seating bowl, each covering 8–12 rows of seating, fed from a common head-end so that the baseband pool is shared. Because the bowl is a confined volume with a 30–40 m radius, each node serves 800–1,000 attendees at 0.15 Erlang, i.e. 120–150 Erlangs, which requires 2.6 GHz channels with 60° sectorisation and 30° adaptive beams.
- **Tier 2 (outdoor small cells).** Six small-cell sites around the concourse and gates, planned at N = 3 with six sectors, carrying the arriving and departing crowd and the standing areas.
- **Tier 3 (macro overlay).** The existing township macro site nearest the stadium, retained for coverage continuity on the approach roads and for the queue outside the gates, and for fallback if a small cell fails.

12.5 Spectrum and Interference Strategy

Dynamic spectrum borrowing. During the event window (16:00–23:00), 40 channels are reallocated from the low-traffic residential and open-ground zones to the stadium cluster, raising the stadium's

allocation from 125 to 165 channels. At N = 7 this gives 23 channels per cell and 18.080 Erlangs per cell at 5 % GoS, a **45 % increase** over the 12.461 Erlangs available from 17 channels. The borrowing is temporary, is logged, and is reversed at the end of the event; the residential zones accept a temporary degradation from 2 % to about 3 % blocking, which is acceptable for a two-day window.

Carrier aggregation and dual connectivity. The 1800 MHz and 2.6 GHz carriers are aggregated at the device, so a single user session draws capacity from both layers. The macro layer provides the control plane and mobility anchor; the 2.6 GHz layer provides the throughput.

30° adaptive beamforming. Twelve-sector narrow-beam arrays at 2.6 GHz divide the bowl into twelve isolated azimuth zones. The beamwidth reduction from 60° to 30° halves the number of first-tier interferers seen by any user and raises the worst-case S/I by roughly 3 dB, which is the margin needed to make the N = 3 plan of Option C viable. Beam steering also tracks crowd movement between sessions, concentrating capacity where the audience actually is.

Massive MIMO spatial multiplexing. A 64T64R array at 3.5 GHz can serve 8–16 users simultaneously on the same time-frequency resource through multi-user spatial multiplexing. In a stadium bowl, where users are separated in elevation as well as azimuth, the spatial separation is unusually favourable, and the practical multiplexing gain of 6–8 layers per cell is the single largest capacity lever available. It is a multiplexing gain, not a reuse gain, so it does not consume interference margin.

Wi-Fi 6E offload. Twenty-four 6 GHz access points carry best-effort video uploads and social media, removing 40–50 % of the data volume from the cellular layers. Offload is the cheapest capacity in the design because the 6 GHz band is licence-exempt and the access points can be deployed in days.

12.6 Relaxed Grade of Service and Traffic Differentiation

The event network deliberately runs at a 5 % GoS for voice rather than 2 %. The justification is that a 5 % blocking probability during a two-hour event is a far smaller social cost than the capital cost of the 17 % additional capacity needed to reach 2 %, and that most event traffic is data, for which blocking is the wrong metric: data sessions queue (Erlang-C behaviour) and a short delay is imperceptible.

The traffic is therefore differentiated into three classes with separate admission policies:

1. **Conversational (VoLTE, video call).** Admitted only if a channel is free; 5 % blocking target; highest scheduling priority.
2. **Streaming (live video, social upload).** Admitted with queueing; a 2 s queue tolerance; scheduled on the 2.6 GHz and 6 GHz layers.
3. **Best effort (messaging, background sync).** Offloaded to Wi-Fi where possible; otherwise admitted last and throttled under congestion.

This class-based admission is what allows the same physical network to deliver an acceptable experience at 5 % voice blocking rather than the 1 % that a naive single-class design would demand.

12.7 Deployment Plan

Table 12.3: Four-day deployment plan

Day	Activity	Deliverable	Acceptance criterion
1	Site survey, RF prediction, spectrum borrowing request, Wi-Fi backhaul survey	RF plan, lease list, borrowing approval	Predicted coverage ≥ −95 dBm in 95 % of bowl
2	Install 6 small cells, 23 pico-nodes, 24 Wi-Fi APs, fibre and power	Physical installation complete	All nodes powered and reachable
3	Integration, parameter set (N = 3, 30° beams, handover, admission control), drive/bench test	Parameter baseline, test report	Throughput ≥ 2 Mbps per user at 80 % load
4	Load test with 500 synthetic users, optimisation, go-live rehearsal	Final parameter set, runbook	Blocking < 5 % voice, < 2 s data delay at peak

12.8 Risk Register and Validation Metrics

Table 12.4: Principal risks and mitigations

Risk	Likelihood	Impact	Mitigation
Spectrum borrowing refused	Medium	High	Pre-negotiate 20-channel fallback; rely on 2.6 GHz overlay
Crowd density exceeds 25,000	Medium	Medium	Admission control; Wi-Fi offload expansion; Tier 3 macro absorbs spill
Backhaul congestion	High	High	Dedicated 10 G fibre per small cell; microwave backup
Interference from adjacent cells at N = 3	Medium	High	30° beams, downtilt, power control; fallback to N = 4
Handover storm at gates	High	Medium	Hysteresis 6 dB, dwell 5 s, gate-area overlapping coverage
Equipment failure during event	Low	High	N + 1 redundancy on pico head-end; macro fallback

The event network is validated against four metrics: voice blocking below 5 % during the peak hour; 90th-percentile data throughput above 2 Mbps per user; handover success rate above 98 %; and call drop rate below 2 %. These are the metrics that determine whether the deployment is judged a success, and they are deliberately different from the township metrics because the traffic mix and the duration are different.

Conclusion

This report has designed, analysed and optimised a hexagonal cellular network for a 120 km² university township and stress-tested it against a high-density event. The principal results are as follows.

An aggregate busy-hour load of 8,000 Erlangs, derived from 80,000 subscribers each generating 0.1 Erlang, requires a cluster size of N = 7 to satisfy an 18 dB co-channel protection ratio at a path-loss exponent of 4. That cluster size yields 17 traffic channels per cell, 10.656 Erlangs of Erlang-B capacity at 2 % blocking, and 756 deployed cells of 247.2 m radius with a reuse distance of 1,132.7 m. The geometry is internally consistent: the lattice-derived reuse distance, the closed-form $Q = \sqrt{21}$, and the area-derived radius all agree to five significant figures.

The interference analysis demonstrates why nominal design is insufficient. The equal-distance estimate of 18.66 dB degrades to 17.82 dB at the exact cell vertex and to 16.73 dB under the conservative two-dominant-interferer bound, so a network planned exactly at the threshold has no margin at the boundary. Sectoring supplies that margin — 23.43 dB nominal and 20.79 dB worst case for three sectors — but at a 53 % loss of trunking efficiency; the capacity is only recovered by spending the margin on a tighter reuse pattern, and N = 4 with three sectors yields 15.252 Erlangs per cell and 525 cells, a 30.6 % reduction in infrastructure.

The link budget proves the network to be capacity-limited rather than coverage-limited: the COST-231 Hata model permits a 2.43 km radius on the downlink and 1.27 km on the uplink, while capacity forces 247 m. The 35 dB of excess margin at the cell edge should therefore be converted into interference reduction through power control rather than radiated at full power.

The dynamic allocation study quantifies the value of adaptivity. Fixed allocation blocks 49.6 % of calls in a hot cell even when the surrounding cluster is half empty; hybrid allocation with a borrowing headroom of two reduces network-average blocking from 0.1634 to 0.1013 under heavy hotspot loading, recovering two-thirds of the pooling bound at a small fraction of its signalling cost.

Finally, the stadium challenge shows the limits of uniform design. At 7,500 Erlangs/km² — 112 times the township density — the baseline plan would need 352 cells of 23 m radius. A dual-band, sectorised, three-tier architecture reduces this to 29 sites by combining a 100 MHz overlay, N = 3 reuse with 30° adaptive beams, dynamic spectrum borrowing worth 45 % additional capacity, Wi-Fi offload, and a deliberate relaxation of the voice GoS to 5 %.

The design has three limitations that should be stated plainly. First, the propagation model is a single-slope empirical fit valid for distances of 1–20 km; it is applied here at 0.25 km, outside its validity range, and the micro-cell and indoor layers would require a dual-slope or ray-tracing model for reliable prediction. Second, the interference analysis assumes a uniform hexagonal grid and constant transmitted power, whereas real networks have irregular site placement, sector imbalance and power control, all of which change the interference distribution; the statistical treatment of shadowing (Section 7.6) partially compensates but does not replace a Monte Carlo simulation over a real clutter database. Third, the traffic model is stationary Poisson, whereas event traffic is strongly non-stationary, with arrival rates varying by an order of magnitude between the opening ceremony and the closing session.

The natural extensions of this work are a Monte Carlo simulation of the interference environment with real terrain and clutter data, an optimisation of the frequency plan by graph colouring rather than by the cyclic stride-7 assignment, and an extension of the traffic model to self-similar packet arrivals for the data-dominant 5G case, where Erlang-B is the wrong tool and queueing analysis of the air interface is required.

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Appendix A: Formula Summary

| Quantity | Formula | Section |

| Traffic intensity | $a = \lambda H$ | 0.1 |

| Aggregate load | $A_{load} = N_{users} \cdot a$ | 0.1 |

| Valid cluster sizes | $N = i^2 + ij + j^2$ | 0.3 |

| Cell area | $A = (3\sqrt{3}/2)R^2$ | 0.3 |

| Reuse distance and ratio | $D = R\sqrt{(3N)}$, $Q = \sqrt{(3N)}$ | 0.3 |

| Nominal S/I | $S/I = (3N)^{(\gamma/2)} / i_0$ | 1.3 |

| Erlang-B | $P_b = (A^C/C!) / \Sigma(A^k/k!)$ | 2.2 |

| Erlang-B recursion | $B(C) = A \cdot B(C-1)/(C + A \cdot B(C-1))$ | 2.2 |

| Channels per cell | $k = \lfloor S/N \rfloor$ | 2.1 |

| Cell count | $N_{cells} = \lceil A_{load}/A_{cell} \rceil$ | 2.4 |

| Radius from area | $R = \sqrt{(A_{cell}/2.5981)}$ | 2.5 |

| Exact vertex S/I | $S/I = 1/\Sigma D_i^{(-\gamma)}$, $D_i \in \{Q \pm 1, \sqrt{(Q^2 \mp 1 \mp Q)}\}$ | 3.2 |

| Conservative bound | $S/I = 1/[2(Q-1)^{(-\gamma)} + (i_0-2)Q^{(-\gamma)}]$ | 3.3 |

| Sectoring gain | $\Delta S/I = 10 \log_{10}(i_{0,omni} / i_{0,sector})$ | 4.1 |

| Cell-splitting power | $P_t' = P_t(R'/R)^{\gamma}$ | 5.3 |

| ACI budget | $(C/I_{adj})_{min} = 18 - A_f$ | 6.1 |

| Noise floor | $N = -174 + 10 \log_{10}(B)$ | 7.2 |

| MAPL | $MAPL = EIRP - P_{rx,min} + G_{rx} - L_{body} - M_{shadow}$ | 7.2 |

| Shadow margin | $M = z\sigma$ | 7.6 |

| COST-231 Hata | $PL = 46.3 + 33.9\log f - 13.82\log h_b - a(h_m) + (44.9 - 6.55\log h_b)\log d + C_m$ | 7.1 |

| Hysteresis distance ratio | $d_{ratio} = 10^{(\Delta H/35.225)}$ | 9.3 |

| Dwell time | $T = 2R/v$ | 9.4 |

| Handover rate | $R_{HO} = N_{calls} \cdot v/(2R)$ | 9.4 |

Appendix B: Assumption Register

| Assumption | Value adopted | Justification | Sensitivity |

| Duplex channel interpretation | 125 duplex channels from 25 MHz | Standard course convention; 25 MHz / 200 kHz | S = 62 triples the site count (Section 10.5) |

| Path-loss exponent | $\gamma = 4.0$ in the core | Built-up campus, BS above roof line | N = 19 at $\gamma = 3.2$ (Table 10.1) |

| Traffic model | Poisson arrivals, exponential holding | Standard Erlang-B assumptions | Measured blocking within 1 % |

| Blocked calls cleared | No queueing for voice | Correct for circuit voice | Data modelled by Erlang-C (Section 2.6) |

| Uniform subscriber distribution | 8,000 E spread evenly over 120 km² | No survey data available | DCA analysis (Task 8) covers non-uniformity |

| Base-station height | 30 m | Typical rooftop/lamppost mast | ±10 m changes PL by 1.8 dB |

| Mobile height | 1.5 m | Handset at head height | a(h_m) ≈ 0 dB at 1800 MHz |

| Shadowing σ | 8 dB | Typical urban/suburban | 10 dB lowers MAPL by 2.1 dB |

| Antenna gain and feeder loss | 15 dBi, 3 dB | 65°/7° panel, 30 m feeder | ±2 dB changes MAPL by 2 dB |

| Interference protection | 18 dB | Given in the assignment brief | 12 dB would permit N = 4 |

MARK DISTRIBUTION

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